



A Functional-Coefficient VAR Model for Dynamic Quantiles and Its Application to Constructing Nonparametric Financial Network

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ABSTRACT

In this article, we investigate a functional coefficient vector autoregressive model for conditional quantiles, in which the interdependences among tail risks such as Value-at-Risk are allowed to vary smoothly with a variable of general economy. Methodologically, we develop an easy-to-implement two-stage procedure to estimate functionals in the dynamic network system based on the deep learning method of neural networks and the local linear smoothing technique. We establish the consistency and the asymptotic normality of the proposed estimator under geometrically β -mixing time series settings. The simulation studies are conducted to show that our new methods work fairly well. The potential of the proposed estimation procedures is demonstrated by an empirical study of constructing and estimating a new type of nonparametric dynamic financial network.

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1. Introduction

Since the seminal work by Koenker and Bassett (1978), quantile regression, also called conditional quantile or regression quantile or dynamic quantile, has become an increasingly popular tool for risk analysis in many fields in economics such as labor economics, macroeconomics and financial risk management. It is well known that quantile regression is concerned with estimating a collection of conditional quantiles over the entire conditional distribution, instead of investigating the conditional mean function of dependent variable. The reader is referred to the review papers by Koenker (2005) and Koenker et al. (2017) for more applications of quantile regression.

In the recent two decades, a great deal of attention has been paid to modeling the tail dependence in the financial statistics/econometrics literature. Among the toolkits of quantile methods, dynamic quantile models are naturally suitable for capturing tail dependence. For example, White, Kim, and Manganelli (2015) proposed an innovative method to directly estimate the sensitivity of Value-at-Risk (VaR) of a given financial institution to shocks to the whole financial system by constructing a vector autoregressive (VAR) model for dynamic quantiles, while Härdle, Wang, and Yu (2016) developed a model to describe the network relationship among VaRs of financial institutions by a flexible nonparametric quantile model with L_1 -penalty. Recently, Zhu et al. (2019) constructed a quantile autoregressive model that embeds the observed dependency structure in a dynamic network.

A first important question arising in modeling dynamic quantiles is how to capture the nonlinearities of tail dependence. To the best of our knowledge, much of the existing literature

assumed constant tail dependence in their models. However, numerous studies have documented time-varying risk interdependence in financial time series and discussed their possible origins and relation to risk spillover; see, for example, Billio et al. (2012), Diebold and Yilmaz (2014), Härdle, Wang, and Yu (2016), Yang and Zhou (2017), Liu, Ji, and Fan (2017), Ando and Bai (2020), and Chen et al. (2025), and references therein. The driving force for the variations of risk interdependence may be the institutional changes or the policy interventions, such as the changes of exchange rate systems and the U.S. quantitative easing policy. A second crucial issue is how to model the dependence among conditional quantiles of financial return distributions. Indeed, since the VaR processes are witnessed to be significantly autocorrelated and interdependent with each other by Engle and Manganelli (2004) and White, Kim, and Manganelli (2015), some important types of tail dependence (e.g., the co-dependence between VaRs) can be excluded if a model does not allow for interdependence among conditional quantiles.

To deal with the aforementioned issues in a simultaneous way, we propose a nonparametric approach involving multivariate dynamic quantile models with nonlinear structures. Different from previous studies, we allow the coefficients of the multivariate dynamic quantile models to vary with a smoothing variable, which is chosen based on certain economic theory or some data-driven methods. In addition, the model allows the latent quantiles of several random variables to depend on lagged quantiles and other lagged covariates, which forms a VAR model with unobserved autoregressors. Thus, this model is termed as a functional-coefficient VAR model for dynamic quantiles

(FCVAR-DQ) and is presented in (1). Since the coefficients of dynamic quantile models play an important role in reflecting the interdependences among dynamic quantiles, under our model setup, one can easily illustrate the variation of tail dependence and its relation with the variable of interest.

With coefficients being functionals and autoregressors being latent conditional quantiles, the proposed FCVAR-DQ model is highly nonlinear and conventional techniques of nonlinear quantile regression are not directly applicable. Indeed, as pointed out by Xiao and Koenker (2009), estimation of specific class of fixed-coefficient conditional autoregressive value at risk (CaViaR) models, proposed by Engle and Manganelli (2004) and further studied by Xiao and Koenker (2009), involves fitting a restricted nonlinear quantile regression, which is complicated both computationally and theoretically. The estimation of FCVAR-DQ model makes this problem become more challenging, since the FCVAR-DQ model nests the fixed-coefficient CaViaR model as a special case and all coefficients in our model are allowed to vary with a smoothing variable. To overcome these difficulties, we develop a nonparametric two-stage procedure to estimate functional coefficients in the FCVAR-DQ model. In the first step, latent conditional quantiles are approximated via a sparsely connected neural network. In the second stage, we perform locally weighted estimation for functional coefficients using the estimated conditional quantiles as autoregressors.

Our contributions to the literature can be summarized as follows. First, the model setting in (1) is general enough to nest many well-known dynamic quantile models in the literature; see, for example, the CaViaR model, the threshold CaViaR model as in Gerlach, Chen, and Chan (2011), and the static VAR for VaR model constructed by White, Kim, and Manganelli (2015). Second, by allowing the coefficients to vary with a smoothing variable, a FCVAR-DQ model provides a new tool to estimate the relationship between the time-varying interdependence of risk and the state variable of economy. Third, a new and simple-to-implement estimation procedure is developed for estimating the proposed quantile model with highly nonlinear structure and latent covariates. In addition, large sample theories for the proposed estimator are established for time series context to construct confidence intervals for functional coefficients in the empirical study. Finally, our empirical analysis provides some strong evidence to show that interdependences among VaRs of four world major financial indices' return series vary significantly with the change of U.S. dollar index. This empirical finding is new to the literature and cannot be revealed from using a standard constant-coefficient model.¹

The rest of this article is organized as follows. In Section 2, the model setup and the two-stage estimation procedure are

presented for the FCVAR-DQ model. In addition, large sample theories for the proposed estimator are investigated in this section too, together with constructing a consistent estimator of the asymptotic covariance matrix and corresponding confidence interval. A Monte Carlo simulation study is conducted in Section 3 to illustrate the finite sample performance of the proposed model and its estimation procedure. In Section 4, our proposed model and its modeling procedure are applied to constructing a novel class of nonparametric financial networks based on the real example. Finally, a conclusion is given in Section 5, and the descriptions of some important notations, assumptions, and mathematical proofs are gathered in the online appendix.

2. FCVAR Model for Dynamic Quantiles

2.1. Model Setup

Let Y_{it} ($1 \leq i \leq \kappa$, $1 \leq t \leq n$), a scalar dependent variable, be the i th observation at time t for fixed κ , $\mathcal{F}_{t-1}$ represent the information set up to time $t-1$ for all individuals $1 \leq i \leq \kappa$, and $q_{\tau,t,i}$ be the τ th conditional quantile of Y_{it} given $\mathcal{F}_{t-1}$. Then, we study the following functional-coefficient VAR model for dynamic quantiles, termed as FCVAR-DQ model,

$$q_{\tau,t,i} = \gamma_{i0,\tau}(Z_{it}) + \sum_{s=1}^q \gamma_{i,s,\tau}^\top(Z_{it}) \mathbf{q}_{\tau,t-s} + \sum_{l=1}^p \beta_{i,l,\tau}^\top(Z_{it}) \mathbb{Y}_{t-l} \quad (1)$$

for some p and q , where $\mathbf{q}_{\tau,t} = (q_{\tau,t,1}, \dots, q_{\tau,t,\kappa})^\top$ is a $\kappa \times 1$ vector of conditional quantiles at time t , and $\mathbb{Y}_t$ is a $\kappa_1 \times 1$ vector of covariates with fixed κ_1 , including possibly some or all of $\{Y_{it}\}_{i=1}^\kappa$ and/or some exogenous information $\{x_{it}\}$. In addition, $\gamma_{i0,\tau}(\cdot)$ is a scalar function and is allowed to depend on τ , both $\gamma_{i,s,\tau}(\cdot) = (\gamma_{si1,\tau}(\cdot), \dots, \gamma_{sik,\tau}(\cdot))^\top$ and $\beta_{i,l,\tau}(\cdot) = (\beta_{i1l,\tau}(\cdot), \dots, \beta_{i\kappa_1 l,\tau}(\cdot))^\top$ are $\kappa \times 1$ and $\kappa_1 \times 1$ vectors of functional coefficients, respectively, and they are allowed to depend on τ too. Here, Z_{it} is an observable scalar smoothing variable, which might be one part of $\mathbb{Y}_{t-l}$ and/or other exogenous variables $\{x_{it}\}$ or their lagged variables. Of course, Z_{it} can be an economic index to characterize economic activities. Also, note that Z_{it} can be set as a multivariate variable. In such a case, the estimation procedures and the related theory for the univariate case still hold for multivariate case, but more complicated notations are involved and models with Z_{it} in very high dimension are often not practically useful due to the "curse of dimensionality". In addition, note that both κ and κ_1 in model (1) are assumed to be fixed, which might be relatively strong settings.² It is also noted that similar to the setting of the multi-quantile CaViaR model as in White, Kim, and Manganelli (2008), one may further generalize model (1) by allowing τ in $q_{\tau,t,i}$ to vary across different equations, only with mild changes

¹We found that Chen et al. (2025) considered a time-varying network based on the VAR model with both the coefficient and (error) precision matrices evolving smoothly over time. Different from their motivations, we aim to explore the varying interdependence among lagged conditional quantiles and focus on estimating a dynamic quantile model with functional coefficients and latent covariates. In addition, since the coefficients in our model vary with a smoothing variable, the coefficients of the recursive form of our model (1) are the product of increasing number of functionals, which can not be approximated by a functional at time t . Hence, our estimation strategies are also different from that for the locally stationary models in Chen et al. (2025).

²Indeed, in our empirical study, it is sufficient to consider VAR model with finite dimensions in the sense that we are interested in measuring tail dependence among four major financial market indices. If both κ and κ_1 (or one of them) are allowed to diverge (a function of sample size and going to infinity when sample size goes to infinity), then, model (1) becomes a high-dimensional VAR model, which involves completely new theories and is out of the scope of this article.

on the asymptotic theory in this article. Thus, in order to meet our empirical motivation, all of τ 's in model (1) are the same throughout this article.

Importantly, when estimating dynamic financial network in our empirical studies, we follow White, Kim, and Manganeli (2015) and consider only the tail dependence between current state and the state of one-period lagged, and take $\mathbb{Y}_t$ to be $\mathbb{Y}_t = (|Y_{1t}|, \dots, |Y_{\kappa t}|)^\top$ with $|\cdot|$ representing absolute value.³ Furthermore, the smoothing variable Z_{it} varies only across different time periods but keeps constant over individual units. Therefore, in this article, for easy exposition, our focus is on the simple case that $q = p = 1$, $\kappa = \kappa_1$, $\mathbb{Y}_t = (|Y_{1t}|, \dots, |Y_{\kappa t}|)^\top$, and $Z_{it} = Z_t$ for all $1 \leq i \leq \kappa$. Then, model (1) can be rewritten as

$$q_{\tau,t,i} = \mathbf{g}_{i,\tau}^\top(Z_t) \mathbf{X}_t, \quad (2)$$

where $\mathbf{g}_{i,\tau}(\cdot) = (\gamma_{i0,\tau}(\cdot), \gamma_{i1,\tau}(\cdot), \dots, \gamma_{i\kappa,\tau}(\cdot), \beta_{i1,\tau}(\cdot), \dots, \beta_{i\kappa,\tau}(\cdot))^\top$ is a $(2\kappa + 1) \times 1$ vector of functional coefficients and $\mathbf{X}_t = (1, q_{\tau,t-1,1}, \dots, q_{\tau,t-1,\kappa}, |Y_{1(t-1)}|, \dots, |Y_{\kappa(t-1)}|)^\top$. As discussed in Engle and Manganeli (2004), taking absolute value on lagged Y_{it} allows for illustrating the case when both very positive and negative returns increase the VaR in the future.

It is worthwhile to note that if $q_{\tau,t,i}$ in model (2) is defined as VaR of return Y_{it} , then, $\{\gamma_{ij,\tau}(Z_t)\}_{i=1,j=1}^\kappa$ in model (2) becomes the dependence of VaR for one portfolio at time t on that of another at time $t - 1$. With these functional coefficients $\{\gamma_{ij,\tau}(Z_t)\}_{i=1,j=1}^\kappa$, define the following $\kappa \times \kappa$ matrix

$$\mathbf{\Gamma}_{1,\tau}(Z_t) = (\gamma_{ij,\tau}(Z_t))_{\kappa \times \kappa}, \quad (3)$$

which can be used to characterize the cross-sectional dependence among conditional quantiles $\{q_{\tau,t,j}\}_{j=1}^\kappa$ at time t .⁴ Then, (2) can be expressed as a matrix form, which is a FCVAR model for $q_{\tau,t}$ with exogenous variables,

$$q_{\tau,t} = \boldsymbol{\gamma}_{0,\tau}(Z_t) + \mathbf{\Gamma}_{1,\tau}(Z_t) q_{\tau,t-1} + \mathbf{\Gamma}_{\beta,1,\tau}(Z_t) \mathbb{Y}_{t-1}, \quad (4)$$

where $\boldsymbol{\gamma}_{0,\tau}(Z_t)$ and $\mathbf{\Gamma}_{\beta,1,\tau}(Z_t)$ are defined obviously. Therefore, $\mathbf{\Gamma}_{1,\tau}(Z_t)$ in (4) can serve as a dynamic network system changing with both τ and some information variable Z_t , and it is in a nonparametric nature, so that it is a nonparametric dynamic network. Note that the general setting in the dynamic network system (4) covers some famous network models for characterizing financial risk system, including the one formed by VAR for VaR model in White, Kim, and Manganeli (2015), which assumes the tail dependence $\{\gamma_{ij,\tau}(Z_t)\}_{i=1,j=1}^\kappa$ to be constant and the static financial network in Adrian and Brunnermeier (2016) and Härdle, Wang, and Yu (2016) as special cases.

To investigate the large sample behavior of the proposed estimator (see Theorems 1 and 2), it is assumed throughout this article that the process $\{(Y_{it}, Z_t)\}$ in model (2) is strictly stationary and geometrically β -mixing (absolutely regular).⁵

Indeed, in Appendix D, we show that under some regularity conditions, the joint process $\{(Y_{it}, x_{it}, Z_t, q_{\tau,t,i})\}$ generated by model (1) is strictly stationary and β -mixing with exponentially decaying rate (defined in Appendix D), which is a special case of geometrically β -mixing. By letting $p = q = 1$, these sufficient conditions for the process $\{(Y_{it}, x_{it}, Z_t, q_{\tau,t,i})\}$ in model (1) can be directly applied for verifying the aforementioned probabilistic properties of $\{(Y_{it}, Z_t)\}$ in model (2). The derivation of these two properties in this article is of independent interest, since our main interests are to derive the asymptotic theory for model (2) and to estimate a new class of dynamic financial networks. Therefore, we provide some sufficient conditions that imply these important probabilistic properties and corresponding rigorously theoretical justifications in Appendix D.

Before introducing our estimation procedure in Section 2.2, it should be noted that model (2) can be transformed into a functional-coefficient quantile function, which can serve as an approximation for latent $q_{\tau,t,i}$ in (2) in the preliminary step of our estimation. To see this, we first focus our attention on (4), which is a vector form of (2). Under Assumptions A.1 and A.6 given in Appendix A, by recursively substituting in (4), one has

$$q_{\tau,t} = \mathbf{\Gamma}_{0,\tau,t} + \sum_{l=1}^{\infty} \mathbf{\Gamma}_{l,\tau}(Z_{t,l}) \mathbb{Y}_{t-l}, \quad (5)$$

where $\mathbf{\Gamma}_{0,\tau,t} = \sum_{l=1}^{\infty} \mathbf{\Gamma}_{l0,\tau}(Z_{t,l})$ and $Z_{t,l} = (Z_t, Z_{t-1}, \dots, Z_{t-l+1})^\top$ is a vector of lagged smoothing variables. Here, $\mathbf{\Gamma}_{10,\tau}(Z_{t,1}) = \boldsymbol{\gamma}_{0,\tau}(Z_t)$ and $\mathbf{\Gamma}_{l0,\tau}(Z_{t,l}) = \prod_{\ell=0}^{l-2} \mathbf{\Gamma}_{1,\tau}(Z_{t-\ell}) \boldsymbol{\gamma}_{0,\tau}(Z_{t-l+1})$ for $l \geq 2$; $\mathbf{\Gamma}_{1,\tau}(Z_{t,1}) = \mathbf{\Gamma}_{\beta,1,\tau}(Z_t)$ and $\mathbf{\Gamma}_{l,\tau}(Z_{t,l}) = \prod_{\ell=0}^{l-2} \mathbf{\Gamma}_{1,\tau}(Z_{t-\ell}) \mathbf{\Gamma}_{\beta,1,\tau}(Z_{t-l+1})$ for $l \geq 2$. Now, let $\alpha_{li0,\tau}(Z_{t,l}) = \boldsymbol{\gamma}_{i1,\tau}^\top(Z_t) (\prod_{\ell=1}^{l-2} \mathbf{\Gamma}_{1,\tau}(Z_{t-\ell}) \boldsymbol{\gamma}_{0,\tau}(Z_{t-l+1}))$ for $l \geq 3$, $\alpha_{2i0,\tau}(Z_{t,2}) = \boldsymbol{\gamma}_{i1,\tau}^\top(Z_t) \boldsymbol{\gamma}_{0,\tau}(Z_{t-1})$, and $\alpha_{li0,\tau}(Z_{t,1}) = \gamma_{i0,\tau}(Z_t)$; let $\alpha_{lij,\tau}(Z_{t,l}) = \boldsymbol{\gamma}_{i1,\tau}^\top(Z_t) (\prod_{\ell=1}^{l-2} \mathbf{\Gamma}_{1,\tau}(Z_{t-\ell}) \mathbf{\Gamma}_{\beta,1j,\tau}(Z_{t-l+1}))$ for $l \geq 3$, $\alpha_{2ij,\tau}(Z_{t,2}) = \boldsymbol{\gamma}_{i1,\tau}^\top(Z_t) \mathbf{\Gamma}_{\beta,1j,\tau}(Z_{t-1})$ and $\alpha_{1ij,\tau}(Z_{t,1}) = \beta_{ij,\tau}(Z_t)$. Here, $\gamma_{i0,\tau}(\cdot)$ and $\beta_{ij,\tau}(\cdot)$ are defined in (2), $\boldsymbol{\gamma}_{i1,\tau}^\top(\cdot)$ is the i th row of $\mathbf{\Gamma}_{1,\tau}(\cdot)$, $\mathbf{\Gamma}_{\beta,1j,\tau}(\cdot)$ is the j th column of $\mathbf{\Gamma}_{\beta,1,\tau}(\cdot)$. Then, $\alpha_{li0,\tau}(\cdot)$ and $\alpha_{lij,\tau}(\cdot) = (\alpha_{li1,\tau}(\cdot), \dots, \alpha_{lik,\tau}(\cdot))$ are the i th row of $\mathbf{\Gamma}_{l0,\tau}(\cdot)$ and $\mathbf{\Gamma}_{l,\tau}(\cdot)$, respectively. Thus, the i th row of (5) is written as

$$q_{\tau,t,i} = \sum_{l=1}^{\infty} \alpha_{li0,\tau}(Z_{t,l}) + \sum_{l=1}^{\infty} \alpha_{li,\tau}^\top(Z_{t,l}) \mathbb{Y}_{t-l}. \quad (6)$$

As discussed in Section 2.2, in the first stage of our estimation procedure, a truncated version of (6) is used to approximate latent $q_{\tau,t,i}$ and is modeled globally by a neural network method.

Note that (6) corresponds to a functional-coefficient quantile autoregressive (AR) process of infinity order, which can be used as an underlying data generating process for Y_{it} in model (1). Indeed, denote U_{it} ($1 \leq i \leq \kappa$, $1 \leq t \leq n$) as iid standard uniform random variables on the set of $[0, 1]$. Then, following the same argument as in Koenker and Xiao (2006), by assuming that the right side of (6) is monotonically increasing in τ , model (6) corresponds to a functional-coefficient quantile AR(∞) process as follows

$$Y_{it} = \sum_{l=1}^{\infty} \alpha_{li0}(U_{it}, Z_{t,l}) + \sum_{l=1}^{\infty} \alpha_{li}^\top(U_{it}, Z_{t,l}) \mathbb{Y}_{t-l} \quad (7)$$

³Of course, one can take $\mathbb{Y}_t$ as any other covariates.

⁴It is well known that it is not easy to characterize the cross-sectional dependence among individual conditional quantiles $\{q_{\tau,t,j}\}_{j=1}^\kappa$ at time t for given τ . Therefore, $\mathbf{\Gamma}_{1,\tau}(Z_t)$ defined in (3) can be used as one of measures to capture the dependence.

⁵If some regressors or some of $\{q_{\tau,t,i}\}$ are nonstationary, say, unit root or local to unit root, the model in (1) and its asymptotic theory become very complicated and it needs definitely a further investigation, warranted as a future research; see, for example, Cheng, Han, and Inoue (2022) for details.

where $\alpha_{li0}(\cdot, \cdot)$ is a scalar and measurable function of U_{it} and $\mathbf{Z}_{t,i}$ (from $\mathbb{R} \times \mathbb{R}^l$ to $\mathbb{R}$) and $\alpha_{li}(\cdot, \cdot)$ is a vector of measurable functions from $\mathbb{R} \times \mathbb{R}^l$ to $\mathbb{R}$. Since our objectives in this article are estimating model (2) and modeling the time-varying tail dependence among dynamic quantiles, we only present identification assumptions for (2) and (4) in Appendix A, instead of focusing on model (7) with more complex coefficients. Therefore, further investigating the properties of (7) is left for future research. Next, we make some remarks on our model in (1).

Remark 1. (Special Cases) The proposed FCVAR-DQ model (1) is related to the papers by Engle and Manganelli (2004) and Xiao and Koenker (2009), which discuss the relation between modeling the dynamic structures of conditional quantiles and conditional volatility of returns. Indeed, if $\kappa = \kappa_1$ in (1), Y_{it} in (1) takes a simple form as $Y_{it} = \sigma_{it} e_{it}$, where σ_{it}^2 is the conditional variance of Y_{it} and e_{it} is an iid sequence of random variables with mean zero and unit variance, then, $q_{\tau,t,i} = \sigma_{it} F_e^{-1}(\tau)$, where $F_e(\cdot)$ is the distribution function of e_{it} . Furthermore, if $Y_{it} = \sigma_{it} e_{it}$ is generated from a functional coefficient multivariate GARCH (p, q)-type process for κ ($\kappa \geq 1$) returns extended from the setting in Taylor (1986) as follows

$$\sigma_{it} = \gamma_{i0}(Z_t) + \sum_{s=1}^q \gamma_{is}^\top(Z_t) \Sigma_{t-s} + \sum_{l=1}^p \beta_{il}^\top(Z_t) \mathbb{Y}_{t-l},$$

where $\Sigma_t = (\sigma_{it}, \dots, \sigma_{\kappa t})^\top$ and $\mathbb{Y}_t = (|Y_{1t}|, \dots, |Y_{\kappa t}|)^\top$, then, model (1) reduces to following dynamic quantile model:

$$q_{\tau,t,i} = \gamma_{i0,\tau}(Z_t) + \sum_{s=1}^q \gamma_{is,\tau}^\top(Z_t) q_{\tau,t-s} + \sum_{l=1}^p \beta_{il,\tau}^\top(Z_t) \mathbb{Y}_{t-l}, \quad (8)$$

where $\gamma_{i0,\tau}(\cdot) = \gamma_{i0}(\cdot) F_e^{-1}(\tau)$, $\gamma_{is,\tau}(\cdot) = (\gamma_{si1}(\cdot), \dots, \gamma_{sik}(\cdot))^\top$ and $\beta_{il,\tau}(\cdot) = (\beta_{li1,\tau}(\cdot), \dots, \beta_{lik,\tau}(\cdot))^\top$ with $\beta_{lij,\tau}(\cdot) = \beta_{lij}(\cdot) F_e^{-1}(\tau)$. Note that if γ' 's and β' 's in (8) are constant, model (8) reduces to that in Engle and Manganelli (2004) and Xiao and Koenker (2009), respectively. For details, the reader is referred to the aforementioned papers. Finally, note that if $q_{\tau,t}$ would be observable and all coefficients are threshold functions, model (1) covers the model in Tsay (1998).

Remark 2. (Monotonicity) The issue of monotonicity is frequently discussed for the quantile autoregression model. A specific case for the monotonicity of (1) to hold is that $\{\gamma_{is,\tau}(Z_t)\}_{i=1,s=1}^{\kappa,q}$ and $\{\beta_{il,\tau}(Z_t)\}_{i=1,l=1}^{\kappa,p}$ are all monotone increasing functions with respect to τ , and $\mathbb{Y}_t$ is a positive random vector. In other cases, the assumption of monotonicity can be satisfied by conducting certain data transformation techniques; see Koenker and Xiao (2006) and Fan and Fan (2006) for detailed discussions.

Remark 3. Let $\kappa = \kappa_1$ and $Z_{it} = Z_t$. Then, the matrix form of model (1) is given by $q_{\tau,t} = \gamma_{0,\tau}(Z_t) + \sum_{s=1}^q \Gamma_{s,\tau}(Z_t) q_{\tau,t-s} + \sum_{l=1}^p \Gamma_{\beta,l,\tau}(Z_t) \mathbb{Y}_{t-l}$, where $\Gamma_{s,\tau}(Z_t)$ and $\Gamma_{\beta,l,\tau}(Z_t)$ are defined obviously. Note that for $1 \leq t \leq n$, $1 \leq s \leq q$ and $1 \leq l \leq p$, if all eigenvalues of $\Gamma_{s,\tau}(Z_t)$ and $\Gamma_{\beta,l,\tau}(Z_t)$ have modulus less than 1, then, the matrix form of model (1) corresponds to a recursive form similar to (5). Thus, under this additional assumption, the proposed estimation procedure and theories for model (2) are

extendable to model (1), with more complicated notations being involved.

2.2. Two-Stage Estimation Procedure

In this article, we only focus on estimating the functional coefficients $g_{i,\tau}(\cdot)$ in (2) for simplicity, rather than jointly forecasting $q_{\tau,t,i}$ or doing impulse response analysis. So, it is sufficient to estimate $g_{i,\tau}(\cdot)$ in an equation-by-equation way for each i . Thus, with a little abuse of notation, i is dropped in what follows.

By Assumption A.1 in Appendix A, $\alpha_{l0,\tau}(\cdot)$ and each entry of $\alpha_{l,\tau}(\cdot) = (\alpha_{l1,\tau}(\cdot), \dots, \alpha_{lk,\tau}(\cdot))^\top$ defined in (6) decrease at a geometric rate; that is, there exist positive constants $\rho < 1$ and c , such that $\max_{1 \leq l \leq n} |\alpha_{lj,\tau}(Z_{t,l})| \leq c\rho^l$ for $j = 0, \dots, \kappa$. Since $\alpha_{lj,\tau}(\cdot)$ decreases geometrically, by choosing truncation parameter $m_n = m(n) = m$, (6) becomes

$$\begin{aligned} q_{\tau,t} &= \sum_{l=1}^{m_n} \alpha_{l0,\tau}(Z_{t,l}) + \sum_{l=1}^{m_n} \alpha_{l,\tau}^\top(Z_{t,l}) \mathbb{Y}_{t-l} \\ &\equiv \sum_{l=1}^{m_n} \alpha_{1,l,\tau}^\top(Z_{t,l}) \mathbf{W}_{t-l} \equiv \alpha_\tau^\top(Z_t) \mathbb{W}_t \equiv q_\tau(\mathbf{V}_t), \end{aligned} \quad (9)$$

where $\alpha_{1,l,\tau}(\cdot) = (\alpha_{l0,\tau}(\cdot), \alpha_{l,\tau}^\top(\cdot))^\top = (\alpha_{l0,\tau}(\cdot), \alpha_{l1,\tau}(\cdot), \dots, \alpha_{lk,\tau}(\cdot))^\top$, $\mathbf{W}_t = (1, \mathbb{Y}_t^\top)^\top$, $\alpha_\tau(Z_t) = (\alpha_{1,1,\tau}(Z_{t,1}), \dots, \alpha_{1,m,\tau}(Z_{t,m}))^\top$, $\mathbf{Z}_t = \mathbf{Z}_{t,m}$ by the definition of $\mathbf{Z}_{t,m}$, $\mathbb{W}_t = (\mathbf{W}_{t-1}^\top, \dots, \mathbf{W}_{t-m}^\top)^\top$ and $\mathbf{V}_t = (\mathbf{Z}_t^\top, \mathbb{W}_t^\top)^\top$.

Now, we introduce some preliminary settings of our neural networks. For a vector $\mathbf{v} = (v_1, \dots, v_J)^\top$, denote $\sigma(\mathbf{v}) = (\sigma(v_1), \dots, \sigma(v_J))^\top$ as a vector of activation functions. In this article, we use the rectified linear unit (ReLU) activation function $\sigma(z) = \max(z, 0)$, due to its reliable performance and computational efficiency. See Krizhevsky, Sutskever, and Hinton (2012) for more discussions. In addition, for some integer $L_n \equiv L \geq 2$, let $\mathbf{p} = (p_0, \dots, p_L)^\top \in \mathbb{N}^{L+1}$. We call L the depth of the neural network. Furthermore, let $\tilde{W}_k$ and $\tilde{b}_k$ be a $p_k \times p_{k-1}$ weighted matrix and p_k -dimensional column vector, respectively. Then, a L -layer neural network with input dimension p_0 and output dimension p_L is a function $m: \mathbb{R}^{p_0} \rightarrow \mathbb{R}^{p_L}$ that satisfies:

$$\begin{aligned} m(z) &= \tilde{W}_L m_{L-1}(z) + \tilde{b}_L, m_{L-1}(z) \\ &= \sigma(\tilde{W}_{L-1} m_{L-2}(z) + \tilde{b}_{L-1}), \dots, m_1(z) \\ &= \sigma(\tilde{W}_1 z + \tilde{b}_1), \end{aligned} \quad (10)$$

where m_k is the k th hidden layer and p_k is called the width of the k th layer, for $1 \leq k \leq L-1$. Let $W_k = (\tilde{W}_k, \tilde{b}_k) \in \mathbb{R}^{p_k \times (p_{k-1}+1)}$, we can rewrite (10) as

$$m(z) = W_L \tilde{\sigma} \circ \dots \circ W_2 \tilde{\sigma}(W_1 \tilde{z}), \quad (11)$$

where $\tilde{\sigma}(\mathbf{v}) = (\sigma(\mathbf{v}), 1)^\top$ and $\tilde{z} = (z^\top, 1)^\top$. Following the lead of Schmidt-Hieber (2020), for $R_n \equiv R \in \mathbb{N}$, $L \geq 2$ and fixed $F >$

0, we consider a neural network class with sparse connections as follows:

$$\begin{aligned} \mathcal{M}(R, L, \mathbf{p}, F) = \left\{ m(z) = W_L \tilde{\sigma} \circ \cdots \circ W_2 \tilde{\sigma}(W_1 \tilde{z}) \mid \right. \\ \left. \begin{aligned} &W_k \in \mathbb{R}^{p_k \times (p_{k-1} + 1)}, \\ &\|W_k\|_\infty \leq 1 \text{ for } k = 1, \dots, L, \\ &\sum_{k=1}^L \|W_k\|_0 \leq R \text{ and } \|m\|_\infty \leq F \end{aligned} \right\}. \quad (12) \end{aligned}$$

Remark 4. (Neural networks and functionals with product structure) As shown in (6), each functional coefficient in (9) is the product of increasing number of functionals. Thus, estimating (9) using either kernel or tensor-product sieve methods can be undermined by the extreme curse of dimensionality. In addition, applying deep learning methods to estimating functionals with product structure may lead to overfitting, since the total number of parameters in (10), which is $\sum_{k=1}^L p_k(p_{k-1} + 1)$, can be very large. To overcome these problems, we follow the strategy of Schmidt-Hieber (2020) by imposing sparsity into the connection of neurons between the adjacent hidden layers. As pointed out by Zhong and Wang (2024), the proposed neural networks are able to automatically detect the intrinsic structure of nonparametric model and enjoy a convergence rate that does not depend on the dimension of covariates and smoothing variables.

Step 1: In the first stage, we simplify the notation and use $\mathcal{M}_F$ to denote $\mathcal{M}(R, L, \mathbf{p}, F)$ in (12), with $p_0 = m(\kappa + 1)$ and $p_L = 1$. For the choice of the other tuning parameters, see Section 2.5. Hence, $\hat{q} = \operatorname{argmin}_{q \in \mathcal{M}_F} Q_0(q)$, where $Q_0(q)$ is given by

$$Q_0(q) = \frac{1}{n} \sum_{t=m+1}^n \rho_\tau \{Y_t - q_\tau(\mathbf{V}_t)\}, \quad (13)$$

$\rho_\tau(y) = y[\tau - I(y < 0)]$ is called the “check” (loss) function and $I(A)$ is the indicator function of any set A . By minimizing (13), $q_{\tau,t}$ can be estimated, denoted by $\hat{q}_{\tau,t} = \hat{q}_\tau(\mathbf{V}_t)$.

Step 2: With $\hat{q}_{\tau,t}$ and $\hat{\mathbf{X}}_t = (1, \hat{q}_{\tau,t-1,1}, \dots, \hat{q}_{\tau,t-1,K}, |Y_{1(t-1)}|, \dots, |Y_{K(t-1)}|)^\top$ in place, $\mathbf{g}_\tau(\cdot)$ in (2) is estimated by a local linear estimation method; see Cai and Xu (2008) for details. In particular, under the smoothness condition on the coefficient functions $\mathbf{g}_\tau(\cdot)$ in Assumption A.3 in Appendix A, for any given grid point $z_0 \in [a, b]$, when Z_t is in a neighborhood of z_0 , $\mathbf{g}_\tau(Z_t)$ can be approximated by a polynomial function as $\mathbf{g}_\tau(Z_t) \approx \sum_{r=0}^S \mathbf{g}_\tau^{(r)}(z_0)(Z_t - z_0)^r / r!$, where $\approx$ denotes the approximation by ignoring the higher order terms and $\mathbf{g}_\tau^{(r)}(\cdot)$ is the r th derivative of $\mathbf{g}_\tau(\cdot)$. Hence, we can minimize the following locally weighted loss function $Q_1(\Theta)$ at any given grid point $z_0 \in [a, b]$ to obtain the local linear estimate $\hat{\Theta}$, where

$$Q_1(\Theta) = \sum_{t=1}^n \rho_\tau \left\{ Y_t - \sum_{r=0}^1 \hat{\mathbf{X}}_t^\top \Theta_{r,\tau} (Z_t - z_0)^r \right\} K_h(Z_t - z_0). \quad (14)$$

Here, $\Theta_{r,\tau} = \mathbf{g}_\tau^{(r)}(\cdot) / r!$, $K(\cdot)$ is a kernel function, $K_h(u) = K(u/h)/h$, and $h = h(n)$ is a bandwidth sequence.

2.3. Large Sample Theory

We study the asymptotic distribution of the proposed nonparametric estimator in this section. In the first stage of our estimation procedure, the quantile functions are approximated by the method of neural networks. Recently, Zhong and Wang (2024) considered neural-network approximation in partially linear quantile regression models with iid observations. To prove Theorem 1 in this article, we follow a similar proof strategy as in Zhong and Wang (2024), but substantially extend their results to allow strictly stationary and geometrically β -mixing data setting. This is new to the literature.

Given a d_0 -tuple nonnegative integers, $\mathbf{w} = (w_1, \dots, w_{d_0})$, set $|\mathbf{w}| = w_1 + \dots + w_{d_0}$ and let $D^\mathbf{w}$ be the differential operator defined by $D^\mathbf{w} = \frac{\partial^{|\mathbf{w}|}}{\partial^{w_1} z_1 \dots \partial^{w_{d_0}} z_{d_0}}$. Let ϱ and B be two positive constants and denote a function $q_\tau : \mathbb{T} \subset \mathbb{R}^{d_0} \rightarrow \mathbb{R}$ as a (ϱ, B) -Hölder smooth function if it satisfies

$$\begin{aligned} \sup_{z \in \mathbb{T}} |D^\mathbf{w} q_\tau(z)| &\leq B \text{ and} \\ \sup_{z, z^* \in \mathbb{T}} |D^\mathbf{w} q_\tau(z) - D^\mathbf{w} q_\tau(z^*)| &\leq B \|z - z^*\|^{\varrho - \lfloor \varrho \rfloor} \text{ for all } |\mathbf{w}| = \lfloor \varrho \rfloor. \end{aligned}$$

Note that in our case, $d_0 = p_0$, where p_0 is the input dimension of the neural network defined in Section 2.2. To simplify notation, recall that $q_{\tau,t} \equiv q_\tau(\mathbf{V}_t)$ defined in (9) and $\hat{q}_{\tau,t} \equiv \hat{q}_\tau(\mathbf{V}_t)$. Now, let $\mathcal{H}_{d_0}^\varrho(\mathbb{T}, B)$ be the class of all such (ϱ, B) -Hölder smooth functions. For fixed $K \in \mathbb{N}$, define a composite function class:

$$\begin{aligned} \mathcal{H}(K, \varrho, \mathbf{d}, \bar{\mathbf{d}}, B) &= \left\{ q_\tau = f_K \circ \dots \circ f_1 : \mathbb{T} \rightarrow \mathbb{R} \mid f_k = (f_{k1}, \dots, f_{kd_k})^\top : \right. \\ &\quad [a_{k-1}, b_{k-1}]^{d_{k-1}} \rightarrow [a_k, b_k]^{d_k} \\ &\quad \left. \text{and } f_{k\tau} \in \mathcal{H}_{d_k}^{\varrho_k}([a_{k-1}, b_{k-1}]^{d_{k-1}}, B) \text{ for some } \right. \\ &\quad |a_{k-1}|, |b_{k-1}|, |a_K|, |b_K| \leq B, \\ &\quad \left. \tau = 1, \dots, d_k \text{ and } k = 1, \dots, K \right\} \equiv \mathcal{H}(B), \quad (15) \end{aligned}$$

where $\varrho = (\varrho_1, \dots, \varrho_K)^\top \in \mathbb{R}_+^K$, $\mathbf{d} = (d_0, d_1, \dots, d_K)^\top \in \mathbb{N}^{K+1}$ and $\bar{\mathbf{d}} = (\bar{d}_1, \dots, \bar{d}_K)^\top \in \mathbb{N}^K$ with $\bar{d}_1 \leq d_0$ and $\bar{d}_k \leq d_{k-1}$, for $k = 2, \dots, K$. Here, $\bar{\mathbf{d}}$ is the intrinsic dimension of the function q_τ in $\mathcal{H}(K, \varrho, \mathbf{d}, \bar{\mathbf{d}}, B)$. In addition, define the effective smoothness indices of function q_τ as $\bar{\varrho} = (\bar{\varrho}_1, \dots, \bar{\varrho}_K)^\top$, with $\bar{\varrho}_k = \varrho_k \prod_{l=k+1}^K (\varrho_l \wedge 1)$, for $k = 1, \dots, K-1$ and $\bar{\varrho}_K = \varrho_K$. To establish the convergence rate in Theorem 1, denote

$$\phi_n = \max_{k \in \{1, \dots, K\}} n^{-\bar{\varrho}_k / (2\bar{\varrho}_k + \bar{d}_k)}. \quad (16)$$

As an example, for $x = (x_1, \dots, x_5)^\top$, consider the function

$$f(x) = f_{31}(f_{21}(f_{11}(x_1), f_{12}(x_2)), f_{22}(f_{13}(x_1, x_3), f_{14}(x_4, x_5))),$$

where all $f_{k\tau}$ are (ϱ, B) -Hölder smooth function with $\varrho \geq 1$. Then, $f \in \mathcal{H}(K, \varrho, \mathbf{d}, \bar{\mathbf{d}}, B)$ with $K = 3$, $\varrho = \bar{\varrho} = (\varrho, \varrho, \varrho)^\top$, $\mathbf{d} = (5, 4, 2, 1)^\top$, $\bar{\mathbf{d}} = (2, 2, 2)^\top$ and $\phi_n = n^{-\varrho / (2\varrho + 2)}$. Finally, let $\mathcal{F}_a^b$ be the σ -algebra generated by $\{(Y_{it}, Z_{it})\}_{t=a}^b$. Then, a

stationary process $\{(Y_{it}, Z_{it})\}_{t=-\infty}^{\infty}$ is said to be geometrically β -mixing process if there exist constants $c > 0$ and $\delta_1 > 0$, such that the mixing coefficient $\beta(n)$ defined by

$$\beta(n) = E \left\{ \sup_{A \in \mathcal{F}_{t+n}^{\infty}} |P(A|\mathcal{F}_{-\infty}^t) - P(A)| \right\}$$

satisfies $\beta(n) \leq 2 \exp(-c \cdot n^{\delta_1})$ for any $n \in \mathbb{N}$.

Remark 5. Notice that (9) is included in the function class $\mathcal{H}(K, \mathbf{q}, \mathbf{d}, \mathbf{d}, B)$. To see this, without loss of generality, we consider the case when $m \geq 3$ and $\kappa = 1$. Similar to the argument in Schmidt-Hieber (2020), we can rewrite (9) as a composition of functions $q_{\tau,t} = f_3 \circ f_2 \circ f_1$, with $f_1 = \left(\mathbb{Y}_{1(t-l)}, \gamma_{0,\tau}(Z_{t-l+1}), \beta_{1,\tau}(Z_{t-l+1}), \gamma_{1,\tau}(Z_{t-l+1}) \right)_{l=1,\dots,m}$, $f_2 = (f_{2r})_{r=1,\dots,2m} = \left(\gamma_{0,\tau}(Z_t), \gamma_{1,\tau}(Z_t) \gamma_{0,\tau}(Z_{t-1}), \dots, \gamma_{1,\tau}(Z_t) \prod_{\ell=1}^{m-2} \gamma_{1,\tau}(Z_{t-\ell}) \gamma_{0,\tau}(Z_{t-m+1}), \beta_{1,\tau}(Z_t) \mathbb{Y}_{1(t-1)}, \gamma_{1,\tau}(Z_t) \beta_{1,\tau}(Z_{t-1}) \mathbb{Y}_{1(t-2)}, \dots, \gamma_{1,\tau}(Z_t) \prod_{\ell=1}^{m-2} \gamma_{1,\tau}(Z_{t-\ell}) \beta_{1,\tau}(Z_{t-m+1}) \mathbb{Y}_{1(t-m)} \right)$, and $f_3(\mathbf{y}) = \sum_{l=1}^{2m} \gamma_l$. Here, $\mathbb{Y}_{1t}$, $\gamma_{0,\tau}(Z_t)$, $\beta_{1,\tau}(Z_t)$ and $\gamma_{1,\tau}(Z_t)$ are the one-dimension version of $\mathbb{Y}_t$, $\gamma_{0,\tau}(Z_t)$, $\Gamma_{\beta,1,\tau}(Z_t)$ and $\Gamma_{1,\tau}(Z_t)$ defined in (4), respectively. Assume that $\gamma_{0,\tau}(\cdot)$, $\beta_{1,\tau}(\cdot)$ and $\gamma_{1,\tau}(\cdot)$ are (ϱ_0, B) -Hölder smooth with $\varrho_0 > 1.5$, and we say a function f is (∞, B) -Hölder smooth if it is (ϱ, B) -Hölder smooth for all $\varrho > 0$. In this case, we have $\bar{\mathbf{q}} = \mathbf{q} = (\varrho_0, \varrho_0(m+1), \infty)^\top$, $\bar{\mathbf{d}} = (1, m+1, 2m)^\top$, $\mathbf{d} = (2m, m(m+2), 2m, 1)^\top$. It can be shown that $\phi_n = n^{-\varrho_0/(2\varrho_0+1)}$, which does not depend on the increasing number of input dimension m .

Theorem 1. Under Assumptions A.1–A.10 given in Appendix A, we have,

$$\lim_{C \rightarrow \infty} \lim_{n \rightarrow \infty} \sup_{q_\tau \in \mathcal{H}(B)} P(\|\hat{q}_\tau - q_\tau\|_{L^2([a,b]^{d_0})} \geq C \phi_n \log^{2+(1/\delta_1)} n) = 0.$$

where δ_1 is a positive constant related to β -mixing coefficient.

Remark 6. Compared to the results of iid case in Zhong and Wang (2024), the rate of convergence in Theorem 1 is slower up to a poly-logarithmic factor $\log^{1/\delta_1}(n)$. Indeed, when the data exhibit strong temporal dependence ($\delta_1 < 1$), it is conceivable that the rate will become slower than that obtained by using data with weak dependence ($\delta_1 \geq 1$).

To state the asymptotic property of $\hat{\mathbf{g}}_\tau(z_0)$, we need to add some notations. Define $\Omega^*(z_0) \equiv E[\mathbf{X}_t \mathbf{X}_t^\top f_{Y|Z,X}(q_\tau(z_0, \mathbf{X}_t)|Z_t, \mathbf{X}_t)|Z_t = z_0]$ with $q_\tau(z_0, \mathbf{X}_t) = \mathbf{g}_\tau^\top(z_0) \mathbf{X}_t$ and $f_{Y|Z,X}(\cdot|Z_t, \mathbf{X}_t)$ denoting the conditional density function of Y_t given Z_t and $\mathbf{X}_t$. In addition, let $\Xi(z_0) \equiv \tau(1-\tau)v_0\Omega(z_0)$, where v_0 is defined in Assumption A.5 in Appendix A and $\Omega(z_0) \equiv E[\mathbf{X}_t \mathbf{X}_t^\top|Z_t = z_0]$. Now, the asymptotic normality of $\hat{\mathbf{g}}_\tau(z_0)$ is presented in the following theorem with its detailed proof relegated to Appendix C.

Theorem 2. Under Assumptions A.1–A.10 provided in Appendix A, we have

$$\sqrt{nh} \left[\hat{\mathbf{g}}_\tau(z_0) - \mathbf{g}_\tau(z_0) - \frac{h^2 \mu_2}{2} \mathbf{g}_\tau^{(2)}(z_0) + o_p(h^2) \right] \xrightarrow{d} \mathcal{N}(0, \Sigma_\tau(z_0)),$$

where $\Sigma_\tau(z_0) = (\Omega^*(z_0))^{-1} \Xi(z_0) (\Omega^*(z_0))^{-1} / f_z(z_0)$ and μ_2 is defined in Assumption A.5 in Appendix A.

Remark 7. Theorem 2 implies that the nonparametric estimator $\hat{\mathbf{g}}_\tau(z_0)$ has the same asymptotic distribution that it would have if $q_{\tau,t-1,i}$ in $\mathbf{X}_t$ is known. Indeed, by Assumption A.8 in Appendix A, it can be shown that the terms in asymptotic variance that involve $\mathbb{W}_t$ in the first step are asymptotically vanishing. In other words, our two-stage estimator $\hat{\mathbf{g}}_\tau(z_0)$ enjoys certain oracle efficiency property as obtained in the nonparametric literature; see, for example, Horowitz and Mammen (2004), Horowitz and Lee (2005), and Ozabaci, Henderson, and Su (2014).

Remark 8. In Assumption A.8 in Appendix A, we provide conditions to imply that $\phi_n \log^{2+(1/\delta_1)}(n) = o((nh)^{-1/2})$. Then, the estimation in Step 1 is under-smoothed and the asymptotic bias from the first-stage estimator becomes negligible. See also Cai (2002) and Cai and Xiao (2012) for more discussions about under-smoothing.

2.4. Covariance Estimate

To construct the confidence intervals for the estimated functional coefficients, we need to obtain consistent estimator of the asymptotic covariance matrix $\Sigma_\tau(z_0)$. To this end, one needs to estimate $\Omega(z_0)$ and $\Omega^*(z_0)$ consistently. The procedure of estimating covariance matrix and constructing confidence interval is summarized as follows:

Step 1: Given any grid point $z_0 \in [a, b]$, use the proposed two-stage procedure to obtain $\hat{\mathbf{X}}_t$ and $\hat{\mathbf{g}}_\tau(z_0)$.

Step 2: Construct estimators $\hat{\Omega}(z_0) = \sum_{t=1}^n \hat{\mathbf{X}}_t \hat{\mathbf{X}}_t^\top K_h(Z_t - z_0)/n$ and $\hat{\Omega}^*(z_0) = \sum_{t=1}^n w_t \hat{\mathbf{X}}_t \hat{\mathbf{X}}_t^\top K_h(Z_t - z_0)/n$, where $w_t = I(\hat{\mathbf{g}}_\tau^\top(z_0) \hat{\mathbf{X}}_t - \delta_n < Y_t \leq \hat{\mathbf{g}}_\tau^\top(z_0) \hat{\mathbf{X}}_t + \delta_n) / (2\delta_n)$ for some $\delta_n \rightarrow 0$ as $n \rightarrow \infty$.

Step 3: Obtain $\hat{\Xi}(z_0) = \tau(1-\tau)v_0\hat{\Omega}(z_0)$ and $\hat{\Sigma}_\tau(z_0) = (\hat{\Omega}^*(z_0))^{-1} \hat{\Xi}(z_0) (\hat{\Omega}^*(z_0))^{-1}$.

Step 4: Let $se(\hat{g}_{j,\tau}(z_0)) = [\widehat{Var}(\hat{g}_{j,\tau}(z_0))/nh]^{1/2}$, where $\widehat{Var}(\hat{g}_{j,\tau}(z_0))$ denotes the j th diagonal element of $\hat{\Sigma}_\tau(z_0)$, with $\hat{g}_{j,\tau}(z_0)$ being the j th element of $\hat{\mathbf{g}}_\tau(z_0)$. Construct the $100(1-\alpha)\%$ confidence interval for $g_{j,\tau}(z_0)$ as $[\hat{g}_{j,\tau}(z_0) - c_{\alpha/2} \times se(\hat{g}_{j,\tau}(z_0)), \hat{g}_{j,\tau}(z_0) + c_{\alpha/2} \times se(\hat{g}_{j,\tau}(z_0))]$, where $c_{\alpha/2}$ is the upper $\alpha/2$ -percentile of standard normal distribution.

Clearly, we can apply under-smoothing to eliminate the asymptotic bias in the second-stage in order to ensure the asymptotic validity of the above confidence interval. In Section C.3 of Appendix C, it shows that the above covariance estimators are consistent: $\hat{\Omega}(z_0) = f_z(z_0)\Omega(z_0) + o_p(1)$ and $\hat{\Omega}^*(z_0) = f_z(z_0)\Omega^*(z_0) + o_p(1)$. The proofs of these results rely on the L_2 -convergence of the estimator $\hat{q}_{\tau,t}$, which is guaranteed by

Theorem 1. Then, $\hat{\Sigma}_\tau(z_0) = (\hat{\Omega}^*(z_0))^{-1} \hat{\Xi}(z_0) (\hat{\Omega}^*(z_0))^{-1}$ is a consistent estimator of $\Sigma_\tau(z_0)$. See Appendix C for details. Note that following the suggestion of the second referee, we also derive the uniform consistency of the second-stage estimator $\hat{g}_\tau(z_0)$, which is presented in Theorem E.1 in Appendix E.

2.5. Practical Implementations

In real applications, we need to know how to choose the smoothing variable Z_t , the tuning parameters in the neural network, the truncation parameter $m = m_n$, and the bandwidth h . To this end, we provide some suggestions below.

First, it is important to choose an appropriate smoothing variable Z_t in applying functional-coefficient VAR model for dynamic quantiles in (2). Knowledge on physical background or economic theory of the data may be very helpful, as we have witnessed in modeling the real data in Section 4 by choosing Z_t to be the first difference of daily log series of the U.S. dollar index. Without any prior information, it is pertinent to choose Z_t in terms of some data-driven methods such as the Akaike information criterion, cross-validation, and some other criteria. Ideally, Z_t can be selected as a linear function of certain explanatory variables according to some optimal statistical selection criterion such as LASSO type methods, or an economic index based on some economic theory; see, for instance, Cai, Juhl, and Yang (2015). Nevertheless, here we would recommend using a simple and practical approach proposed by Cai, Fan, and Yao (2000) or Cai, Juhl, and Yang (2015) in practice.

Second, since the neural network has a layer-by-layer structure, some conventional algorithms for quantile regression (e.g., the interior point algorithm proposed in Koenker and Park (1996)) become infeasible in our implementation. To overcome this issue, we follow Zhong and Wang (2024) and use the *Adam* algorithm as in Kingma and Ba (2014), which is included in the Python package *PyTorch*, see, for instance, Paszke et al. (2019) for details, to solve the optimization problem (13). Due to its computational and memory efficiency, *Adam* algorithm is widely applied in the field of deep learning. In practice, given the neural network in (12), we use the *Adam* algorithm to update the parameters $\{W_1, \dots, W_L\}$, with the initial values $\{W_1^{(0)}, \dots, W_L^{(0)}\}$ being selected as the default values in *PyTorch*.

Third, all tuning parameters in the neural network, such as the depth L , width p , learning rate, mini-batch size and the sparsity parameter R for the neural network class $\mathcal{M}(R, L, p, F)$ in (12), are selected on a hold-out validation dataset. Here, the mini-batch size is defined as the subsample size used to calculate the gradient of the objective function for each iteration. In both the simulation study and the real application, we split the data into a training set (the first $\lfloor a\%n \rfloor$ data) and validation set (the remaining $\lfloor (1 - a\%)n \rfloor$ data), with sample size being n and $a = 80$. In practice, we can use grid search to choose the tuning parameters among a set of candidates by minimizing the check loss on the validation set. The candidates of tuning parameters are gathered in Table A1 in Appendix A.

Fourth, to introduce the sparsity into the neural network, a certain proportion of elements in each row of the weight matrix W_k are enforced to be zero and are kept as zero during the training. Throughout this article, this proportion is set to be

50%. As investigated in Zhong and Wang (2024), a different choice of proportion leads to similar results. After training, if the absolute value of an element of W_k is greater than 1, then, we set it to equal to 1 or -1 , according to the sign of this element.

In addition, under Assumption A.2 in Appendix A, it suffices to select a truncation $m \gtrsim \log n$. In practice, one may apply the forward selection method introduced by Cheng, Honda, and Zhang (2016) to select m . Of course, other data-driven methods such as the Akaike information criterion, cross-validation, and some other criteria can also be considered for selecting m . To reduce computational burden, we choose m as a sufficiently large constant multiple of $n^{1/8}$, which is used in our simulation study in Section 3 and the empirical analysis in Section 4.

Finally, we would like to address how to select the bandwidth h at the second step. It is well known that the bandwidth plays an essential role in the tradeoff between reducing bias and variance. In view of (14), it is about selecting the (under-smoothing) bandwidth in the context of estimating the coefficient functions in the quantile regression. Therefore, we recommend the method proposed in Cai and Xu (2008) for selecting h in (14), which is used in our simulation study in Section 3 and the empirical example in Section 4.

3. A Monte Carlo Simulation Study

In this section, we provide a simulation example to examine the finite sample performance of the proposed two-stage estimation for functional coefficients. In the first stage of estimation, all tuning parameters are selected in the way as discussed in Section 2.5. At Step 2, we choose the bandwidth $\hat{h}$ by the nonparametric Akaike information criterion (AIC) as in Cai and Xu (2008). Finally, the Epanechnikov kernel $K(x) = 0.75(1 - x^2)I(|x| \leq 1)$ is used and $m = \lfloor 0.8n^{1/8} \rfloor$.

In this example, for $1 \leq i \leq 4$, the data generating process (DGP) is given by $Y_{it} = \sigma_{it}\varepsilon_{it}$ with $\sigma_{it} = \gamma_{i0}(Z_t) + \gamma_{i1,\varepsilon_{it}}(Z_t)\sigma_{1(t-1)} + \gamma_{i2,\chi_{it}}(Z_t)\sigma_{2(t-1)} + \gamma_{i3,\varepsilon_{it}}(Z_t)\sigma_{3(t-1)} + \gamma_{i4,\chi_{it}}(Z_t)\sigma_{4(t-1)} + \beta_{i1}(Z_t)|Y_{1(t-1)}| + \beta_{i2}(Z_t)|Y_{2(t-1)}| + \beta_{i3}(Z_t)|Y_{3(t-1)}| + \beta_{i4}(Z_t)|Y_{4(t-1)}|$, where ε_{it} are mutually iid from $\mathcal{N}(0, 1)$, $\gamma_{10}(z) = \gamma_{30}(z) = 1.5 \exp(-3(z+1)^2) + \exp(-8(z-1)^2)$, $\gamma_{20}(z) = \gamma_{40}(z) = 1.5 \exp(-3(z-1)^2) + \exp(-8(z+1)^2)$, $\varepsilon_{it} = 0.2U_{it} + 0.8$ and $\chi_{it} = 0.2 \exp(U_{it}) + 0.8$ with $U_{it} \sim \text{iid Uniform}[0, 1]$. In addition, let $\gamma_{ij,\varepsilon_{it}}(z) = \gamma_{ij}(z) \cdot \varepsilon_{it}$, $\gamma_{ij,\chi_{it}}(z) = \gamma_{ij}(z) \cdot \chi_{it}$. Then, $\gamma_{ij,\varepsilon_{it}}(z)$, $\gamma_{ij,\chi_{it}}(z)$ and $\beta_{ij}(z)$ for $1 \leq i, j \leq 4$ are defined as follows. For $i = 1$, $\gamma_{i1,\varepsilon_{it}}(z) = 0.15 \{1 + \exp(-4z)\}^{-1} \varepsilon_{it}$, $\gamma_{i2,\chi_{it}}(z) = (0.04z^2)\chi_{it}$, $\gamma_{i3,\varepsilon_{it}}(z) = (0.15 \exp(-4z) \{1 + \exp(-4z)\}^{-1})\varepsilon_{it}$, $\gamma_{i4,\chi_{it}}(z) = (0.1 \cos^2(z))\chi_{it}$. For $i = 2$, $\gamma_{i1,\varepsilon_{it}}(z) = \gamma_{i3,\varepsilon_{it}}(z) = (0.1 \sin(-0.8\pi z) + 0.1)\varepsilon_{it}$, $\gamma_{i2,\chi_{it}}(z) = 0.15 \{1 + \exp(-4z)\}^{-1} \chi_{it}$, $\gamma_{i4,\chi_{it}}(z) = (0.15 \exp(-4z) \{1 + \exp(-4z)\}^{-1})\chi_{it}$. For $i = 3$, $\gamma_{i1,\varepsilon_{it}}(z) = \gamma_{i3,\varepsilon_{it}}(z) = (0.1 \sin(0.8\pi z) + 0.1)\varepsilon_{it}$, $\gamma_{i2,\chi_{it}}(z) = \gamma_{i4,\chi_{it}}(z) = (0.1 \cos(0.8\pi z) + 0.1)\chi_{it}$. For $i = 4$, $\gamma_{i1,\varepsilon_{it}}(z) = \gamma_{i3,\varepsilon_{it}}(z) = (0.1 \cos(0.8\pi z) + 0.1)\varepsilon_{it}$, $\gamma_{i2,\chi_{it}}(z) = \gamma_{i4,\chi_{it}}(z) = (0.1 \sin(0.8\pi z) + 0.1)\chi_{it}$. As for $\beta_{ij}(z)$, we set $\beta_{ij}(z) = \gamma_{ij}(z)$ for $1 \leq i, j \leq 4$.

Thus, for $1 \leq i \leq 4$, the quantile function of our DGP is given by the model in (1) with $\kappa = 4$, $\mathbb{Y}_t = (|Y_{1t}|, |Y_{2t}|, |Y_{3t}|, |Y_{4t}|)^T$, $q = p = 1$ and $Z_{it} = Z_t$, where Z_t is generated from Uniform

Table 1. Simulation results for $\gamma_{10,\tau}(\cdot)$, $\gamma_{20,\tau}(\cdot)$, $\gamma_{30,\tau}(\cdot)$, $\gamma_{40,\tau}(\cdot)$, and $\gamma_{ij,\tau}(\cdot)$ for $i = 1, 2$ and for $1 \leq j \leq 4$.

τ	$n = 500$		$n = 1500$		$n = 4000$	
	MADE(γ_{10})	MADE(γ_{20})	MADE(γ_{10})	MADE(γ_{20})	MADE(γ_{10})	MADE(γ_{20})
0.05	0.822 (0.266)	0.989 (0.327)	0.518 (0.114)	0.650 (0.133)	0.429 (0.069)	0.460 (0.078)
0.15	0.543 (0.179)	0.638 (0.223)	0.319 (0.081)	0.367 (0.077)	0.239 (0.048)	0.259 (0.050)
0.95	0.935 (0.332)	1.107 (0.362)	0.551 (0.144)	0.673 (0.151)	0.346 (0.087)	0.472 (0.089)
	MADE(γ_{30})	MADE(γ_{40})	MADE(γ_{30})	MADE(γ_{40})	MADE(γ_{30})	MADE(γ_{40})
0.05	0.911 (0.268)	0.952 (0.353)	0.595 (0.173)	0.688 (0.142)	0.402 (0.098)	0.405 (0.111)
0.15	0.593 (0.199)	0.638 (0.203)	0.370 (0.100)	0.365 (0.124)	0.279 (0.056)	0.270 (0.053)
0.95	1.207 (0.411)	1.153 (0.368)	0.660 (0.183)	0.675 (0.228)	0.418 (0.115)	0.457 (0.127)
	MADE(γ_{11})	MADE(γ_{12})	MADE(γ_{11})	MADE(γ_{12})	MADE(γ_{11})	MADE(γ_{12})
0.05	0.325 (0.156)	0.202 (0.071)	0.257 (0.091)	0.094 (0.033)	0.113 (0.046)	0.062 (0.021)
0.15	0.269 (0.123)	0.218 (0.082)	0.211 (0.078)	0.115 (0.046)	0.107 (0.036)	0.073 (0.026)
0.95	0.366 (0.162)	0.287 (0.106)	0.177 (0.060)	0.125 (0.044)	0.100 (0.031)	0.102 (0.034)
	MADE(γ_{13})	MADE(γ_{14})	MADE(γ_{13})	MADE(γ_{14})	MADE(γ_{13})	MADE(γ_{14})
0.05	0.590 (0.250)	0.234 (0.107)	0.217 (0.073)	0.183 (0.065)	0.090 (0.032)	0.069 (0.027)
0.15	0.284 (0.113)	0.371 (0.162)	0.177 (0.063)	0.168 (0.068)	0.104 (0.036)	0.086 (0.031)
0.95	0.428 (0.165)	0.492 (0.212)	0.153 (0.057)	0.123 (0.038)	0.092 (0.028)	0.109 (0.031)
	MADE(γ_{21})	MADE(γ_{22})	MADE(γ_{21})	MADE(γ_{22})	MADE(γ_{21})	MADE(γ_{22})
0.05	0.421 (0.180)	0.687 (0.317)	0.220 (0.087)	0.299 (0.118)	0.135 (0.051)	0.080 (0.029)
0.15	0.335 (0.130)	0.253 (0.105)	0.224 (0.096)	0.143 (0.057)	0.124 (0.045)	0.088 (0.034)
0.95	0.452 (0.196)	0.334 (0.134)	0.179 (0.076)	0.146 (0.055)	0.108 (0.038)	0.119 (0.040)
	MADE(γ_{23})	MADE(γ_{24})	MADE(γ_{23})	MADE(γ_{24})	MADE(γ_{23})	MADE(γ_{24})
0.05	0.662 (0.299)	0.293 (0.129)	0.255 (0.100)	0.221 (0.085)	0.106 (0.039)	0.085 (0.029)
0.15	0.342 (0.133)	0.447 (0.182)	0.201 (0.074)	0.197 (0.073)	0.119 (0.044)	0.103 (0.037)
0.95	0.515 (0.208)	0.586 (0.228)	0.180 (0.067)	0.148 (0.054)	0.104 (0.036)	0.116 (0.044)

$[-2, 2]$ independently. Also, note that $\gamma_{i0,\tau}(\cdot) = \gamma_{i0}(\cdot)\Phi^{-1}(\tau)$, $\gamma_{i1,\tau}(\cdot) = \gamma_{i1}(\cdot)(0.2\tau^2 + 0.8)$, $\gamma_{i3,\tau}(\cdot) = \gamma_{i3}(\cdot)(0.2\tau^2 + 0.8)$, while $\gamma_{i2,\tau}(\cdot) = \gamma_{i2}(\cdot)(0.2\exp(\tau) + 0.8)$, $\gamma_{i4,\tau}(\cdot) = \gamma_{i4}(\cdot)(0.2\exp(\tau) + 0.8)$ and $\beta_{ij,\tau}(\cdot) = \beta_{ij}(\cdot)\Phi^{-1}(\tau)$ for $1 \leq i, j \leq 4$, with $\Phi(\cdot)$ being the distribution function of the standard normal. Therefore, $\gamma_{i0,\tau}(\cdot)$, $\gamma_{ij,\tau}(\cdot)$, and $\beta_{ij,\tau}(\cdot)$ are functions of τ , suggesting different covariate effects at different levels of τ .

It is worthwhile to note that the DGP used in this Monte Carlo study is a multivariate GARCH model with functional coefficients, which is closely related to our real data example in Section 4 for following reasons. First, as discussed in Remark 1, the conditional quantile function of the GARCH process has a representation of dynamic quantile model, which is naturally suitable to describe the interdependences among conditional VaRs in our new financial network. Second, the types of functional coefficients in this simulation seem to be rich enough to cover different classes of variations of tail dependence in real data applications. Third, since GARCH-type models have been proven to be highly successful in financial applications, data that are generated from GARCH-type models may be appropriate to simulate the real data example in this article. Also note that the settings of functional coefficients in this simulation study are similar to those in Tang et al. (2013) and Sherwood and Wang (2016), which satisfy the smoothness conditions given in Assumption A.8 in Appendix A. See Section A.2 in Appendix A for more discussions.

To assess the finite sample performance of the proposed nonparametric estimators, we use the mean absolute deviation error (MADE) for $\gamma_{i0,\tau}(\cdot)$, $\gamma_{ij,\tau}(\cdot)$ and $\beta_{ij,\tau}(\cdot)$, defined as

$$\text{MADE}(\gamma) = \frac{1}{n_0} \sum_k |\hat{\gamma}_\tau(z_k) - \gamma_\tau(z_k)|, \text{ and}$$

$$\text{MADE}(\beta_{ij,\tau}) = \frac{1}{n_0} \sum_k |\hat{\beta}_{ij,\tau}(z_k) - \beta_{ij,\tau}(z_k)|,$$

where $\gamma_\tau(\cdot)$ can be either $\gamma_{ij,\tau}(\cdot)$ or $\gamma_{i0,\tau}(\cdot)$, both $\hat{\gamma}_\tau(\cdot)$ and $\hat{\beta}_{ij,\tau}(\cdot)$ are local linear quantile estimates of $\gamma_\tau(\cdot)$ and $\beta_{ij,\tau}(\cdot)$, respectively, and $\{z_k = 0.1(k-1) - 1.75 : 1 \leq k \leq n_0 = 36\}$ are the grid points. Also, note that in this example, $q_{\tau,t,i} = \sigma_{it}F_\varepsilon^{-1}(\tau) = 0$ when $\tau = 0.5$, which leads the quantile regression problem to be ill-posed so that the results for $\tau = 0.5$ are omitted. Therefore, we only consider τ 's level to be 0.05, 0.15, 0.85, and 0.95 and the sample sizes are $n = 500, 1500$, and 4000. For each setting, we replicate the simulations 500 times and compute the median and standard deviation (in parentheses) of 500 MADE values. The results are reported in Tables 1–4 only for $\tau = 0.05, 0.15$, and 0.95, respectively, but the results for $\tau = 0.85$ are omitted due to the space limitation, available upon request. One can see clearly from Tables 1–4 that both median and standard deviation of 500 MADE values steadily decrease as the sample size increases for all four values of τ .

Finally, we illustrate the finite sample performance for the consistent covariance estimation given in Section 2.4 via evaluating the pointwise confidence intervals (CI) with the asymptotic bias ignored. To do this, define $\widehat{\text{var}}(\cdot)$ as the asymptotic variance calculated by the estimators presented in Section 2.4. Then, we compute the average of empirical coverage rates (AECR) of the 95% pointwise CIs of $\gamma_{ij,\tau}(\cdot)$ and $\beta_{ij,\tau}(\cdot)$ without the asymptotic bias correction for $1 \leq i, j \leq 4$, defined as,

$$\text{AECR}(\gamma_{ij,\tau}) = \frac{1}{n_0 B} \sum_k \sum_{b=1}^B I_b\{\gamma_{ij,\tau}(z_k) \in \hat{\gamma}_{ij,\tau}(z_k) \pm 1.96 \times \text{se}(\hat{\gamma}_{ij,\tau}(z_k))\},$$

where $\text{se}(\hat{\gamma}_{ij,\tau}(\cdot)) = [\widehat{\text{var}}(\hat{\gamma}_{ij,\tau}(\cdot))/nh]^{1/2}$, $I_b\{\gamma_{ij,\tau}(\cdot) \in \hat{\gamma}_{ij,\tau}(\cdot) \pm 1.96 \times \text{se}(\hat{\gamma}_{ij,\tau}(\cdot))\}$ is an indicator function which equals to 1 if $\gamma_{ij,\tau}(\cdot)$ is covered by the interval $\hat{\gamma}_{ij,\tau}(\cdot) \pm 1.96 \times \text{se}(\hat{\gamma}_{ij,\tau}(\cdot))$ in the b th replication and 0 otherwise, and the number of replications is $B = 500$. Similarly, $\text{AECR}(\beta_{ij,\tau})$, $\text{se}(\hat{\beta}_{ij,\tau}(\cdot))$, and $I_b\{\beta_{ij,\tau}(\cdot) \in \hat{\beta}_{ij,\tau}(\cdot) \pm 1.96 \times \text{se}(\hat{\beta}_{ij,\tau}(\cdot))\}$ can be defined in the same fashion.

Table 2. Simulation results for $\gamma_{ij,\tau}(\cdot)$ for $i = 3, 4$ and for $1 \leq j \leq 4$.

τ	$n = 500$		$n = 1500$		$n = 4000$	
	MADE(γ_{31})	MADE(γ_{32})	MADE(γ_{31})	MADE(γ_{32})	MADE(γ_{31})	MADE(γ_{32})
0.05	0.345 (0.170)	0.613 (0.283)	0.263 (0.079)	0.357 (0.102)	0.142 (0.049)	0.091 (0.030)
0.15	0.289 (0.127)	0.234 (0.105)	0.265 (0.088)	0.159 (0.058)	0.131 (0.045)	0.096 (0.036)
0.95	0.546 (0.186)	0.364 (0.134)	0.221 (0.064)	0.168 (0.053)	0.118 (0.037)	0.132 (0.038)
	MADE(γ_{33})	MADE(γ_{34})	MADE(γ_{33})	MADE(γ_{34})	MADE(γ_{33})	MADE(γ_{34})
0.05	0.598 (0.294)	0.274 (0.110)	0.301 (0.099)	0.251 (0.077)	0.112 (0.037)	0.088 (0.031)
0.15	0.318 (0.138)	0.395 (0.170)	0.220 (0.090)	0.215 (0.076)	0.121 (0.045)	0.104 (0.039)
0.95	0.593 (0.201)	0.703 (0.228)	0.200 (0.069)	0.172 (0.045)	0.115 (0.033)	0.142 (0.044)
	MADE(γ_{41})	MADE(γ_{42})	MADE(γ_{41})	MADE(γ_{42})	MADE(γ_{41})	MADE(γ_{42})
0.05	0.447 (0.174)	0.781 (0.294)	0.213 (0.091)	0.266 (0.126)	0.148 (0.050)	0.084 (0.031)
0.15	0.297 (0.133)	0.265 (0.105)	0.254 (0.092)	0.159 (0.058)	0.133 (0.047)	0.098 (0.038)
0.95	0.484 (0.182)	0.356 (0.133)	0.243 (0.069)	0.191 (0.059)	0.106 (0.040)	0.124 (0.039)
	MADE(γ_{43})	MADE(γ_{44})	MADE(γ_{43})	MADE(γ_{44})	MADE(γ_{43})	MADE(γ_{44})
0.05	0.805 (0.306)	0.326 (0.123)	0.244 (0.098)	0.206 (0.093)	0.118 (0.041)	0.096 (0.033)
0.15	0.320 (0.151)	0.414 (0.189)	0.228 (0.086)	0.216 (0.083)	0.129 (0.045)	0.108 (0.041)
0.95	0.556 (0.196)	0.671 (0.247)	0.240 (0.075)	0.187 (0.055)	0.103 (0.036)	0.122 (0.047)

Table 3. Simulation results for $\beta_{ij,\tau}(\cdot)$ for $i = 1, 2$ and for $1 \leq j \leq 4$.

τ	$n = 500$		$n = 1500$		$n = 4000$	
	MADE(β_{11})	MADE(β_{12})	MADE(β_{11})	MADE(β_{12})	MADE(β_{11})	MADE(β_{12})
0.05	0.234 (0.099)	0.177 (0.084)	0.134 (0.042)	0.109 (0.043)	0.082 (0.033)	0.066 (0.023)
0.15	0.169 (0.063)	0.128 (0.049)	0.089 (0.033)	0.071 (0.030)	0.059 (0.023)	0.053 (0.018)
0.95	0.212 (0.091)	0.177 (0.081)	0.136 (0.047)	0.108 (0.040)	0.094 (0.029)	0.073 (0.024)
	MADE(β_{13})	MADE(β_{14})	MADE(β_{13})	MADE(β_{14})	MADE(β_{13})	MADE(β_{14})
0.05	0.176 (0.083)	0.158 (0.074)	0.119 (0.040)	0.096 (0.033)	0.069 (0.023)	0.063 (0.023)
0.15	0.131 (0.051)	0.120 (0.047)	0.073 (0.029)	0.066 (0.025)	0.048 (0.018)	0.040 (0.016)
0.95	0.190 (0.080)	0.165 (0.072)	0.109 (0.042)	0.096 (0.038)	0.075 (0.025)	0.069 (0.021)
	MADE(β_{21})	MADE(β_{22})	MADE(β_{21})	MADE(β_{22})	MADE(β_{21})	MADE(β_{22})
0.05	0.279 (0.129)	0.210 (0.096)	0.168 (0.073)	0.122 (0.049)	0.115 (0.045)	0.087 (0.034)
0.15	0.191 (0.081)	0.148 (0.060)	0.113 (0.037)	0.083 (0.035)	0.078 (0.031)	0.058 (0.023)
0.95	0.291 (0.120)	0.211 (0.094)	0.163 (0.073)	0.120 (0.052)	0.110 (0.043)	0.078 (0.029)
	MADE(β_{23})	MADE(β_{24})	MADE(β_{23})	MADE(β_{24})	MADE(β_{23})	MADE(β_{24})
0.05	0.242 (0.098)	0.198 (0.094)	0.134 (0.054)	0.119 (0.052)	0.093 (0.030)	0.079 (0.030)
0.15	0.166 (0.067)	0.150 (0.063)	0.087 (0.040)	0.077 (0.032)	0.061 (0.023)	0.057 (0.020)
0.95	0.236 (0.093)	0.203 (0.093)	0.137 (0.053)	0.117 (0.050)	0.094 (0.033)	0.079 (0.033)

Table 4. Simulation results for $\beta_{ij,\tau}(\cdot)$ for $i = 3, 4$ and for $1 \leq j \leq 4$.

τ	$n = 500$		$n = 1500$		$n = 4000$	
	MADE(β_{31})	MADE(β_{32})	MADE(β_{31})	MADE(β_{32})	MADE(β_{31})	MADE(β_{32})
0.05	0.258 (0.101)	0.209 (0.091)	0.183 (0.058)	0.153 (0.053)	0.108 (0.038)	0.094 (0.033)
0.15	0.199 (0.084)	0.146 (0.066)	0.118 (0.044)	0.099 (0.037)	0.076 (0.029)	0.062 (0.024)
0.95	0.312 (0.097)	0.267 (0.087)	0.195 (0.060)	0.146 (0.046)	0.118 (0.039)	0.093 (0.031)
	MADE(β_{33})	MADE(β_{34})	MADE(β_{33})	MADE(β_{34})	MADE(β_{33})	MADE(β_{34})
0.05	0.225 (0.092)	0.206 (0.080)	0.146 (0.049)	0.146 (0.045)	0.085 (0.033)	0.089 (0.031)
0.15	0.156 (0.068)	0.148 (0.063)	0.098 (0.041)	0.094 (0.033)	0.064 (0.022)	0.064 (0.023)
0.95	0.272 (0.082)	0.228 (0.076)	0.163 (0.050)	0.147 (0.047)	0.098 (0.031)	0.089 (0.031)
	MADE(β_{41})	MADE(β_{42})	MADE(β_{41})	MADE(β_{42})	MADE(β_{41})	MADE(β_{42})
0.05	0.296 (0.119)	0.264 (0.104)	0.163 (0.069)	0.144 (0.059)	0.111 (0.045)	0.098 (0.037)
0.15	0.175 (0.077)	0.157 (0.074)	0.121 (0.048)	0.102 (0.036)	0.079 (0.031)	0.066 (0.024)
0.95	0.288 (0.114)	0.260 (0.099)	0.217 (0.059)	0.176 (0.053)	0.117 (0.040)	0.099 (0.037)
	MADE(β_{43})	MADE(β_{44})	MADE(β_{43})	MADE(β_{44})	MADE(β_{43})	MADE(β_{44})
0.05	0.256 (0.103)	0.242 (0.089)	0.141 (0.052)	0.143 (0.056)	0.090 (0.030)	0.092 (0.034)
0.15	0.155 (0.066)	0.152 (0.061)	0.104 (0.044)	0.091 (0.036)	0.067 (0.025)	0.067 (0.022)
0.95	0.255 (0.107)	0.238 (0.084)	0.183 (0.053)	0.175 (0.048)	0.093 (0.034)	0.092 (0.034)

The simulation results are presented in Table 5, for $n = 4000$ and $\tau = 0.05, 0.15$ and 0.95 . From Table 5, one can see basically that for each setting, the AECRs of the 95% pointwise CIs are quite close to the nominal level 95%. In general, the results of this simulated experiment demonstrate that the proposed procedure is reliable and works fairly well.

4. An Empirical Example

4.1. Empirical Models

In this section, the proposed model and estimation method are applied to constructing and estimating a new class of dynamic financial networks in international equity markets. Different

Table 5. Average of empirical coverage rates (AECR) of 95% pointwise confidence intervals for $\gamma_{ij,\tau}(\cdot)$ and $\beta_{ij,\tau}(\cdot)$ without the asymptotic bias correction, for $1 \leq i, j \leq 4$ and $n = 4000$.

τ	Coverage rates of $\hat{\gamma}_{ij,\tau}(\cdot)$				Coverage rates of $\hat{\beta}_{ij,\tau}(\cdot)$		
	$\hat{\gamma}_{11,\tau}$	$\hat{\gamma}_{12,\tau}$	$\hat{\gamma}_{13,\tau}$	$\hat{\gamma}_{14,\tau}$	$\hat{\beta}_{11,\tau}$	$\hat{\beta}_{12,\tau}$	$\hat{\beta}_{13,\tau}$
0.05	0.926	0.908	0.936	0.918	0.920	0.911	0.925
0.15	0.914	0.942	0.948	0.930	0.930	0.928	0.939
0.95	0.902	0.950	0.936	0.926	0.937	0.943	0.952
0.05	$\hat{\gamma}_{21,\tau}$	$\hat{\gamma}_{22,\tau}$	$\hat{\gamma}_{23,\tau}$	$\hat{\gamma}_{24,\tau}$	$\hat{\beta}_{21,\tau}$	$\hat{\beta}_{22,\tau}$	$\hat{\beta}_{23,\tau}$
	0.941	0.918	0.940	0.931	0.923	0.939	0.913
	0.922	0.944	0.954	0.941	0.931	0.943	0.937
0.15	0.918	0.927	0.921	0.925	0.907	0.942	0.920
0.95	0.949	0.913	0.946	0.935	0.924	0.935	0.934
0.05	$\hat{\gamma}_{31,\tau}$	$\hat{\gamma}_{32,\tau}$	$\hat{\gamma}_{33,\tau}$	$\hat{\gamma}_{34,\tau}$	$\hat{\beta}_{31,\tau}$	$\hat{\beta}_{32,\tau}$	$\hat{\beta}_{33,\tau}$
	0.949	0.913	0.946	0.935	0.924	0.935	0.934
	0.927	0.904	0.946	0.931	0.915	0.931	0.930
0.15	0.929	0.940	0.937	0.941	0.931	0.937	0.949
0.95	0.955	0.941	0.943	0.957	0.939	0.930	0.931
0.05	$\hat{\gamma}_{41,\tau}$	$\hat{\gamma}_{42,\tau}$	$\hat{\gamma}_{43,\tau}$	$\hat{\gamma}_{44,\tau}$	$\hat{\beta}_{41,\tau}$	$\hat{\beta}_{42,\tau}$	$\hat{\beta}_{43,\tau}$
	0.955	0.941	0.943	0.957	0.939	0.930	0.931
	0.942	0.927	0.953	0.944	0.924	0.923	0.939
0.15	0.940	0.950	0.954	0.949	0.939	0.941	0.941
0.95							

from the existing literature, the interdependences of this class of networks vary with a smoothing economic variable. Motivated by White, Kim, and Manganelli (2015), we consider the dependence between current and one-day lagged VaR. In particular, we define each linkage in our network as the dependence between the VaR of return of one market index at time t and that of another at time $t - 1$. Therefore, our network can be written as follows:

$$\text{VaR}_{it} = \mathbf{y}_{i,\tau}^\top (Z_{t-1}) \text{VaR}_{t-1}, \quad i = 1, 2, \dots, \kappa, \quad (17)$$

where $\text{VaR}_{t-1} = (\text{VaR}_{1(t-1)}, \dots, \text{VaR}_{\kappa(t-1)})^\top$ is a vector of VaRs for all market index returns at time $t - 1$ and VaR_{it} is the VaR of the i th market index return at time t , which is described as follows $\text{VaR}_{it} = -\inf\{y \in \mathbb{R} : P(Y_{it} > y | \mathcal{F}_{t-1}) \leq 1 - \tau\} = -\inf\{y \in \mathbb{R} : F(y | \mathcal{F}_{t-1}) > \tau\}$ for $i = 1, 2, \dots, \kappa$ at a given $\tau \in (0, 1)$. Here, $\mathcal{F}_{t-1}$ is the information set to present all information of the i th return available at time $t - 1$ and $F(\cdot | \mathcal{F}_{t-1})$ represents the conditional distribution function of Y_{it} given $\mathcal{F}_{t-1}$. In addition, Z_{t-1} is a smoothing economic variable and $\mathbf{y}_{i,\tau}(\cdot) = (\gamma_{i1,\tau}(\cdot), \dots, \gamma_{i\kappa,\tau}(\cdot))^\top$ is a $\kappa \times 1$ vector of functional coefficients. Then, we extract the quantile estimation of the functional coefficients from (17) and construct the matrix $|\hat{\mathbf{\Gamma}}_{1,\tau}(Z_{t-1})| = (|\hat{\gamma}_{ij,\tau}(Z_{t-1})|)_{\kappa \times \kappa}$ as our financial network, in which, $|\hat{\gamma}_{ij,\tau}(Z_{t-1})|$ represents the strength of dependence between the VaR of return for the market index i at time t and that for the index j at time $t - 1$, under the τ th quantile level, and is driven by the smoothing variable Z_{t-1} . Here, taking absolute value on each $\hat{\gamma}_{ij,\tau}(Z_{t-1})$ enables us to calculate and analyze indicators of connectedness (see Section 4.3 for details). Thus, matrix $|\hat{\mathbf{\Gamma}}_{1,\tau}(Z_{t-1})|$ is useful to capture the risk interdependence and how it changes with a smoothing variable Z_{t-1} . Note that the entries in $|\hat{\mathbf{\Gamma}}_{1,\tau}(Z_{t-1})|$ correspond to the absolute value of the estimated values of $\{\gamma_{ij,\tau}(\cdot)\}$ in the network model in (3). Therefore, our two-stage procedures can be applied here for direct estimation of the interdependence among the VaRs of returns for the market indices.

It is necessary to mention that we restrict our attention to the time-varying interdependence between the VaRs of four financial market indices' returns at time t and those at time

$t - 1$,⁶ though Xu et al. (2022) detected significant contemporaneous effect of connected nodes on the conditional quantile function of financial return in the common shareholder network and the headquarter location based network. Different from the predetermined networks studied in Xu et al. (2022), the proposed financial network in our real example is constructed by tail dependence among unobserved VaRs of financial indices' returns, where observable information of relationship among nodes (e.g., binary network data) can hardly be found. Thus, it still remains unclear to us whether the contemporaneous interaction among VaRs of financial indices is significant in our real example or not. A formal study of this important issue is left for future research.

4.2. Data

Our dataset includes the daily series between January 5, 2006 and February 10, 2021 for four major world equity market indices: the U.K. FTSE 100 Index, the Japanese Nikkei 225 Index, the U.S. S&P 500 Composite Index and the Chinese Shanghai Composite Index (SSE). We model the i th index's return series $Y_{it} = 50 \log(\pi_{it}/\pi_{i(t-1)})$, where $i = 1, 2, 3$, and 4 corresponds to the four aforementioned market indices in turn and π_{it} is i th index level at the t th day. The time range of data includes the financial crisis in the U.S. in 2008, the European sovereign debt crisis of 2011–2012, and the COVID-19 pandemic starting from 2019. The daily series of four market indices are downloaded in Yahoo Finance and the estimation sample size is $n = 3254$. Thus, we take $m = \lfloor 0.8n^{1/8} \rfloor$ and the way of selecting tuning parameters for neural networks is described in Section 2.5. Similar to the example in Monte Carlo simulation study, $\{z_k = 0.1(k-1) - 1 : 1 \leq k \leq n_0 = 21\}$ are the grid points, $\hat{h}$ is selected by the nonparametric AIC criterion as in Cai and Xu (2008)

⁶Note that because of volatility persistence, the VaR of financial return may have a long memory behavior. However, according to the empirical results in Engle and Manganelli (2004) and White, Kim, and Manganelli (2015), the specification of autoregressive for dynamic quantile with lag one seems to perform well in out-of-sample testing. Thus, to simplify the implementation of our methodology, we follow White, Kim, and Manganelli (2015) and only consider modeling the dynamic pattern between VaR at time t and its one-period lag in this empirical study.

and the Epanechnikov kernel is used. Although it is feasible to introduce more kinds of market index into the equation system (17), due to the computational burdens, we only consider the risk co-dependences among four major markets' indices.

As for the smoothing variable Z_t , we choose $Z_t = 50 \log(D_t/D_{t-1})$, where D_t is the U.S. dollar index on the t th day and can be downloaded from the Federal Reserve Bank of St. Louis. The U.S. dollar index measures value of U.S. dollar against the currencies of a broad group of major U.S. trading partners, higher values of the index indicate a stronger U.S. dollar. This choice of smoothing variable is reasonable, because the exchange rate has been regarded as an important factor associated with international transmission of risk in many empirical studies. For instance, Menkhoff et al. (2012) discussed the relation between innovations in global foreign exchange volatility and excess returns arising from strategies of carry trade, through which the risk spillover transmits from one country to others. In addition, Yang and Zhou (2017) showed that volatility spillover intensity increases with U.S. dollar depreciation. We do not claim that the U.S. dollar index is the only choice for smoothing variable, but we choose the U.S. dollar index because it contains more information about risk transmission among international equity markets. It is desirable to consider other variables of economic status as the smoothing variable and this may be left in a future study.

4.3. Empirical Results

The empirical analysis in this section includes two steps: First, we estimate $\gamma_{ij,\tau}(Z_{t-1})$ for each market index in the equation system in (17) under $\tau = 0.05$. Second, we use the estimated value of $\gamma_{ij,\tau}(Z_{t-1})$ to construct the matrix $|\hat{\mathbf{\Gamma}}_{1,\tau}(Z_{t-1})|$, and do network analysis based on this matrix.

Before exploring the matrix $|\hat{\mathbf{\Gamma}}_{1,\tau}(Z_{t-1})|$, it is important to examine whether each $\gamma_{ij,\tau}(Z_{t-1})$ in (17) varies significantly with Z_{t-1} or not. To this end, we estimate each $\gamma_{ij,\tau}(Z_{t-1})$ and the corresponding 95% pointwise CIs with the asymptotic bias eliminated via under-smoothing. Figure 1 depicts the corresponding estimation results, in which ij th panel represents the result for $\gamma_{ij,\tau}(\cdot)$, respectively. The black solid line in each panel of Figure 1 represents the estimates of the $\gamma_{ij,\tau}(\cdot)$ for $1 \leq i, j \leq 4$ in (17) along various values of Z_{t-1} under $\tau = 0.05$, and the red dashed lines are 95% pointwise confidence intervals for each estimate. From Figure 1, we clearly see that most of the estimated coefficient functions vary significantly over the interval $[-1, 1]$, which means that we can not use fixed-coefficient dynamic quantile models to fit the data.

Next, we consider analyzing the matrix $|\hat{\mathbf{\Gamma}}_{1,\tau}(Z_{t-1})|$, in which each entry is $|\gamma_{ij,\tau}(Z_{t-1})|$. To simplify notation, Z_{t-1} and τ are dropped from $|\gamma_{ij,\tau}(Z_{t-1})|$ in the matrix $|\hat{\mathbf{\Gamma}}_{1,\tau}(Z_{t-1})|$, in what follows. Then, $|\hat{\gamma}_{ij}|$ in the matrix $|\hat{\mathbf{\Gamma}}_{1,\tau}(Z_{t-1})|$ represents the magnitude of dependence between the risk of market index j at time $t - 1$ and that of market index i at time t . For the purpose of visualization, by following Härdle, Wang, and Yu (2016), we first define the levels of connectedness. The connectedness with respect to the incoming links (CIL) is defined as $\sum_{j=1}^4 |\hat{\gamma}_{ij}|$, which measures the risk spillover that was emitted from all four market indices one day ago and is received currently by

a certain market index. Analogously, the connectedness with respect to the outgoing links (COL) is defined as $\sum_{i=1}^4 |\hat{\gamma}_{ij}|$, which measures the risk spillover emitted from a certain market index one day ago and is received currently by all market indices. Here, $i, j = 1, 2, 3, 4$ correspond to the four aforementioned market indices in turn. Intuitively, the CIL measures exposures of individual indices to systemic shocks from the financial network, while the COL measures contributions of individual indices for risk events in the network. Other than the CIL and COL, we also analyze the total connectedness in the global market, which is equal to $\sum_{j=1}^4 \sum_{i=1}^4 |\hat{\gamma}_{ij}|$ and indicates the total risk spillover in the global market, see Härdle, Wang, and Yu (2016) for more applications about these types of connectedness.

Figures 2 and 3 display the corresponding results along the same values of Z_{t-1} , under $\tau = 0.05$, respectively. In Figure 2, each panel displays the CIL and COL subject to the U.S. dollar change. The solid line in each panel represents values of COL and the dashed line indicates values of the CIL. For Figure 3, the vertical axis measures the total connectedness appeared in international equity markets and the horizontal axes in both figures are the same as those in each panel of Figure 1.

Figure 2 shows that the curves of all four major market indices vary greatly over the interval $[-1, 1]$ and become larger when there is either appreciation ($Z_{t-1} > 0$) or depreciation ($Z_{t-1} < 0$) of U.S. dollar. In particular, when the U.S. dollar experiences appreciation and during the “bad times” of the market (when $Z_{t-1} > 0$ and $\tau = 0.05$), the domestic prices of commodity in Europe, Japan, and China may increase, which pose risks on domestic companies. Then, all investors who invested corporations in the European, Japanese, and Chinese markets suffer from some loss of returns, causing both the CIL and COL to go up in all three markets. For the U.S. market, U.S. assets may become favorable among global investors during the U.S. dollar appreciation, while investors in the U.S. market who invested corporations in the rest of the world face some loss of returns. These two forces lead the U.S. market to be both more important to the global market and to be more vulnerable to risk events in the global market, respectively. Thus, both curves in the panel of S&P 500 index increase.

As for the case when U.S. dollar depreciated ($Z_{t-1} < 0$), profits of investment on domestic corporations in European, Japanese, and Chinese markets may increase, which encourage investors to give leverage in investing corporations in these three markets. Therefore, both types of curves in all three markets, as well as the CIL in the U.S. market increase. Nevertheless, global investors who invested assets in the U.S. market subject to adverse situation, which results in an upward movement of COL of S&P 500 index in the U.S. market.

It is interesting that in the European market, during the U.S. dollar appreciation ($Z_{t-1} > 0$), the COL dominates CIL. These dynamic patterns in the European market may be explained by the so called “carry trade”. The carry trade refers to borrowing a low-yielding asset and buying a higher-yielding foreign asset to earn the interest rate differential plus the expected foreign currency appreciation. Due to the relatively lower interest rate in the European market within our time span of study, as $Z_{t-1} > 0$, carry traders who borrowed low-yielding assets from the European market and bought assets from the U.S. market enjoy

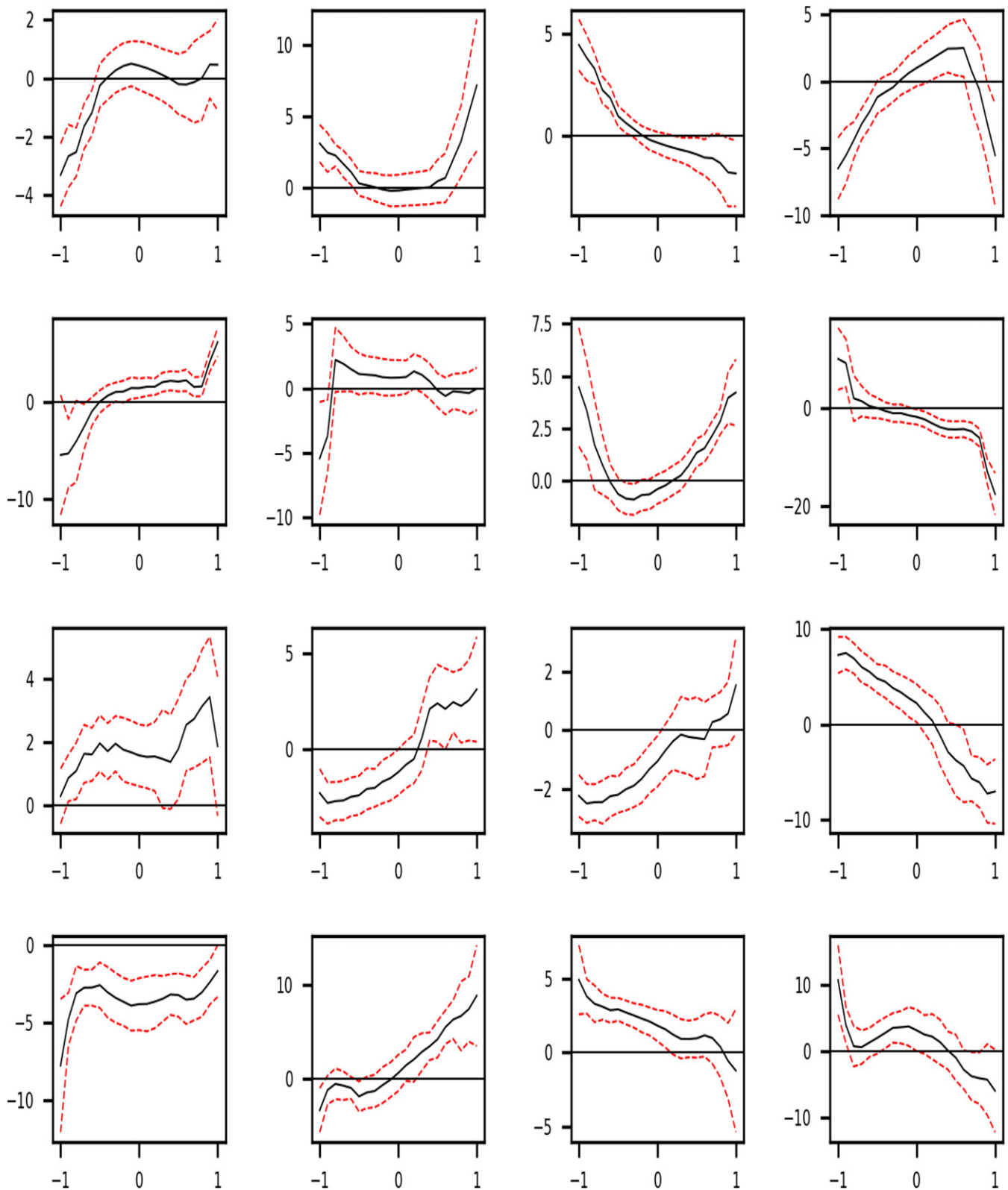


Figure 1. Plots of the estimated coefficient functions $\gamma_{ij,\tau}(\cdot)$ for $1 \leq i \leq 4$ and $1 \leq j \leq 4$ in (17) in the main article under $\tau = 0.05$ (black solid lines), in which j th panel represents the result for $\gamma_{ij,\tau}(\cdot)$, respectively. The red dashed lines in each panel indicate the 95% pointwise confidence interval for the estimate with the asymptotic bias ignored.

the increase of excess returns to carry trade. As a result, the European market becomes less vulnerable to risk events caused by carry traders, which makes the CIL become smaller than COL. While in the U.S. market, since the price of risky assets

relies heavily on the demand of carry trade during U.S. dollar appreciation, it becomes much easier for the U.S. market to be affected by the global market. Therefore, the CIL dominates the COL in the U.S. market.

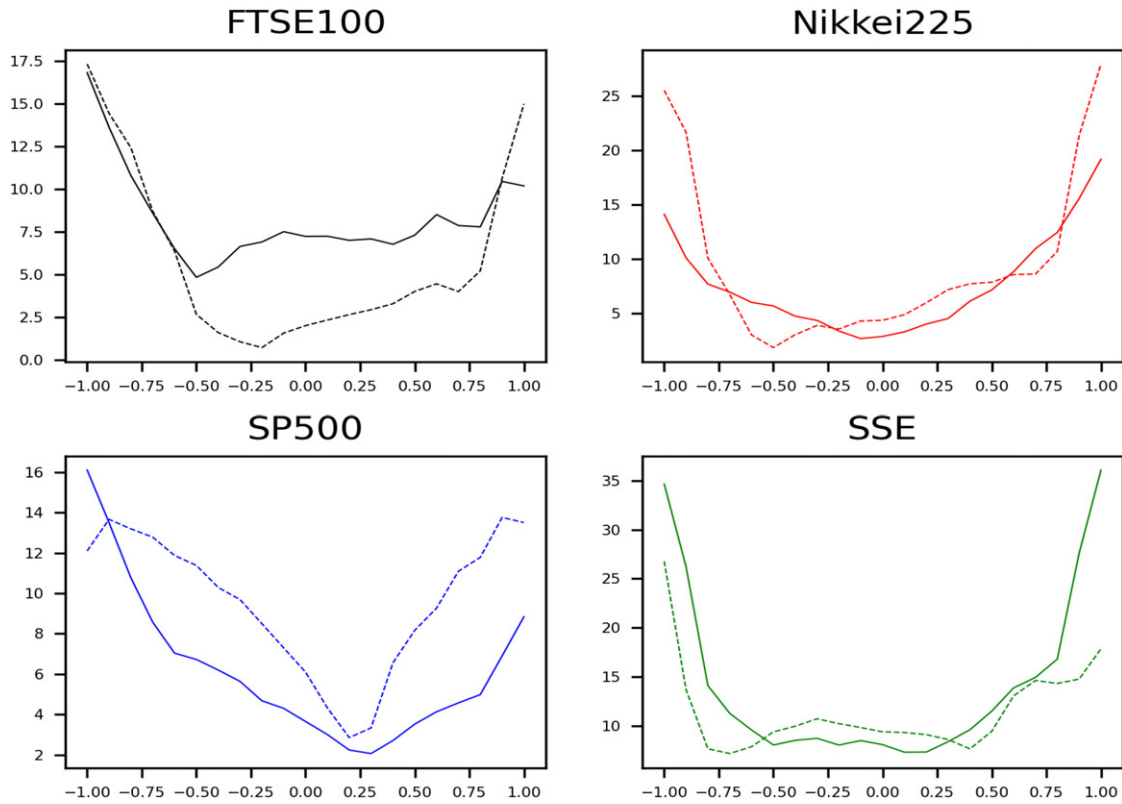


Figure 2. Connectedness with respect to outgoing links and connectedness with respect to incoming links for four market indices with $\tau = 0.05$. The solid line in each panel represents values of connectedness with respect to the outgoing links and the dashed line in each panel indicates values of connectedness is for the incoming link.

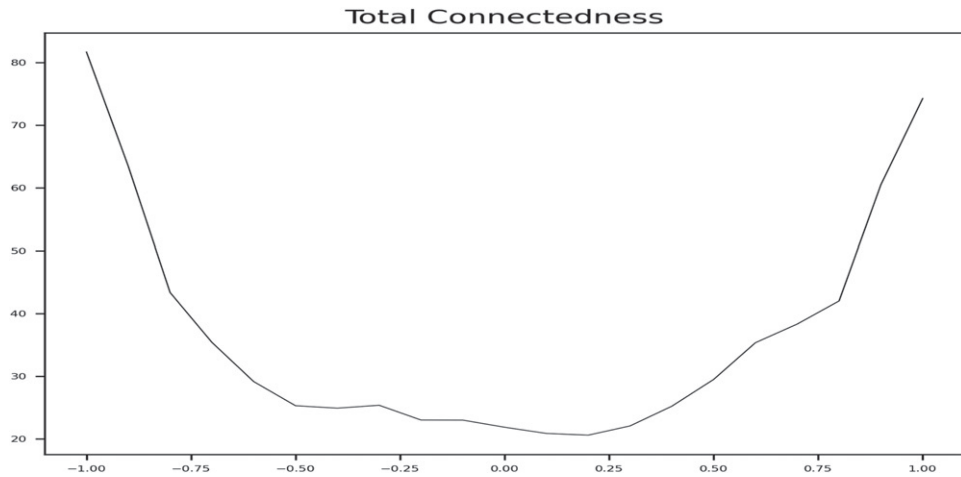


Figure 3. Total connectedness in international equity markets with $\tau = 0.05$.

Figure 3 sheds light on the variation of risk spillover in the global financial market. Observed that in Figure 3, the total connectedness of all four market indices demonstrates an U-shaped pattern. It means that total risk spillover in the four major markets decreases when Z_{t-1} becomes larger within the interval $[-1, 0]$. As Z_{t-1} exceeds 0, the risk spillover intensity is magnified. In general, Figure 3 shows that the relationship between total risk spillover and the U.S. dollar change switches its pattern at a certain threshold of the U.S. dollar change, which is a relatively new result in literature.

Finally, we present two figures in Appendix F for the path of estimated connectedness with respect to the outgoing links (COL), the path of connectedness with respect to the incoming

links (CIL) in Figure A1, and the path of total connectedness in Figure A2, for the case of $\tau = 0.05$. See Appendix F for the details. From Figures A1 and A2, one may see that these dynamic patterns of COL and CIL in the European and the U.S. market may be explained by the carry trade. We conjecture that carry trading may play important role in causing the co-movement of the risk events between the European and the U.S. market.

5. Conclusion

In this article, we investigate a functional coefficient VAR model for conditional quantiles, which is new to the literature. A two-

stage procedure is proposed to estimate the coefficients functionals and the asymptotic normality for the proposed estimator is established. The simulation results show that our new estimation methods work fairly well. In addition, there is little literature regarding the relationship between the variation of financial network and the general state of economy. Our proposed framework allows one to study how the network characteristics of risk spillover in a financial system vary with the state of an economy.

There are several issues still worth of further studies. First, it is interesting to visualize the topological change of our financial network and to measure the transition of risk spillover among different market indices when the general economy is shifting. Technically, these studies can be realized by our econometric model. Second, the asymptotic properties of the functional coefficient estimators in our model provide solid theory to test the abnormal variation of financial network. Third, it is meaningful to allow for cross-sectional dependence in the current model. Although some methods have been developed to deal with cross-sectional dependence in the literature of conditional mean models, due to the nature of conditional quantile model, it is not obvious how to extend these to the quantile setting. Finally, if Z_t in (2) is time, then the model in (2) provides a good start for studying conditional quantile estimation of ARCH- and GARCH-type models with time-varying parameters; see, for example, the papers by Dahlhaus and Subba Rao (2006) and Chen and Hong (2016) for the time-varying GARCH type models. We leave these important issues, together with some possible extensions as mentioned earlier in the article, as future research topics.

Supplementary Materials

The supplementary material file contains mathematical proofs and additional empirical results.

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The authors claim that there are no relevant financial or non-financial competing interests to report for this article. Also, the authors declare that they do not use any generative AI and AI-assisted technologies in the writing process.

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